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V810i S2EX Imaging

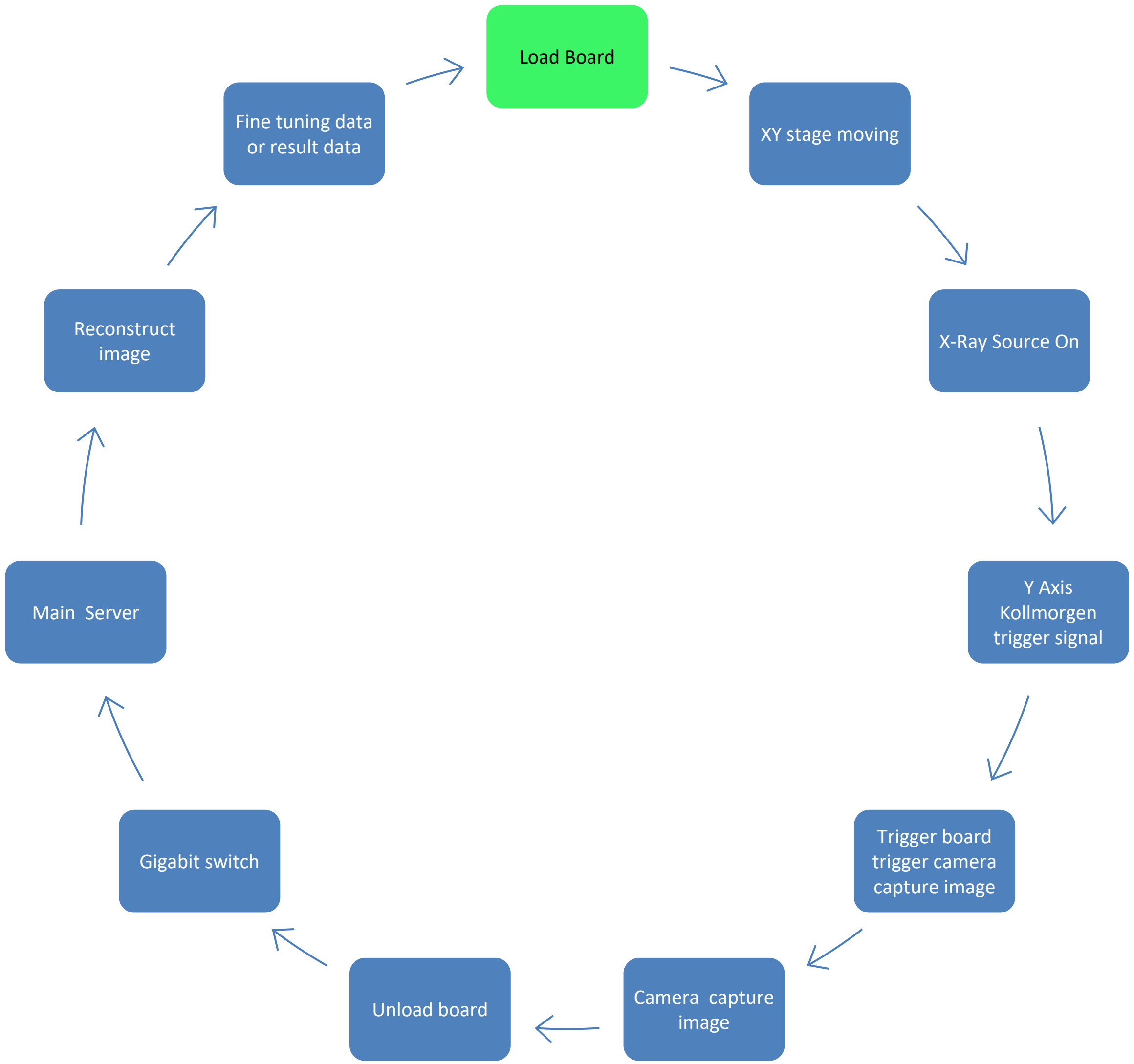
ABI
Automated Board Inspection

Imaging Introduction

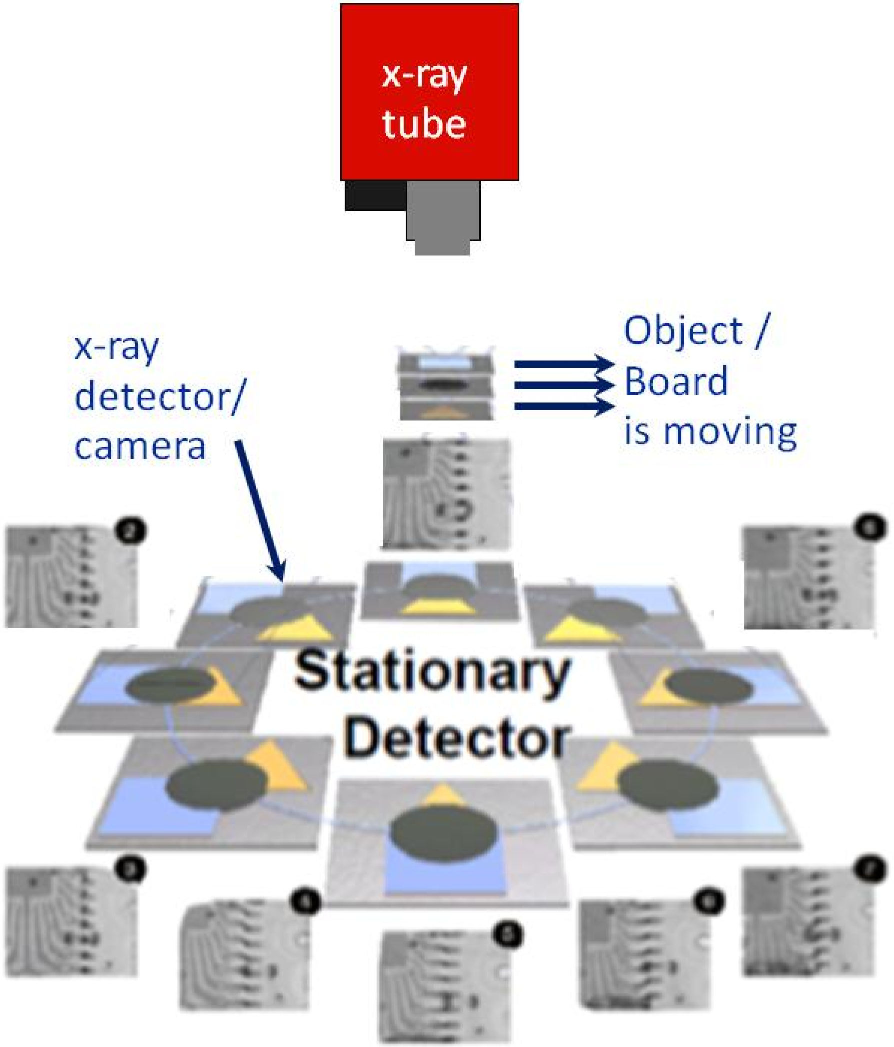
Hardware and software that :

- Generate X-Radiation
- Capture Images
- Process Images

Imaging Process Flow



Tomosynthesis



- As opposed to 5DX’s Laminography, object/board is moving in Tomosynthesis during image acquisition while x-ray spot and detector (camera) are not.
- To obtain different angle images, multiple TDI cameras are installed at different angle with respect to the object. In this example 8 cameras are installed.

Projection Images (Top view)

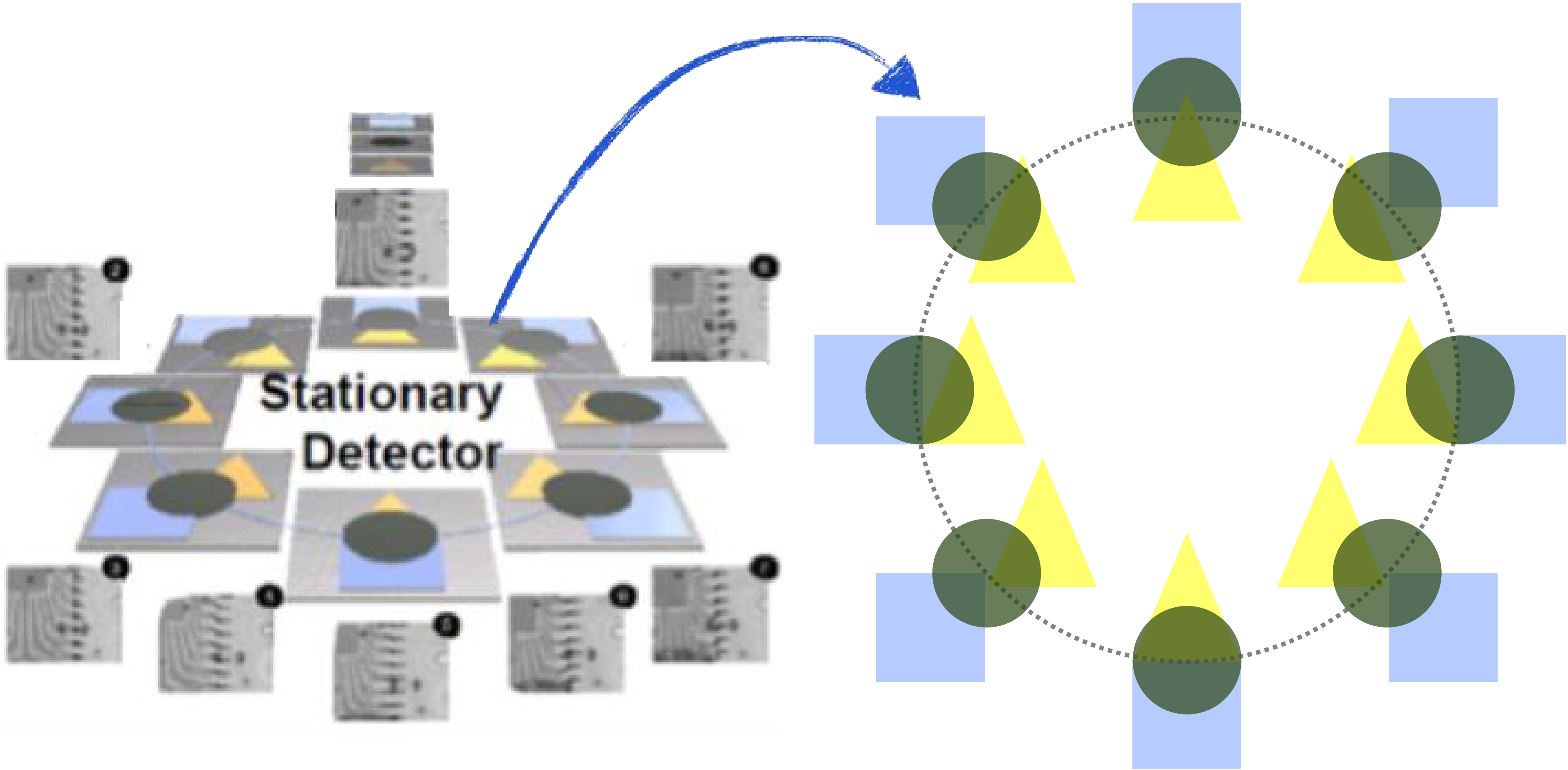


Image Reconstruction Process

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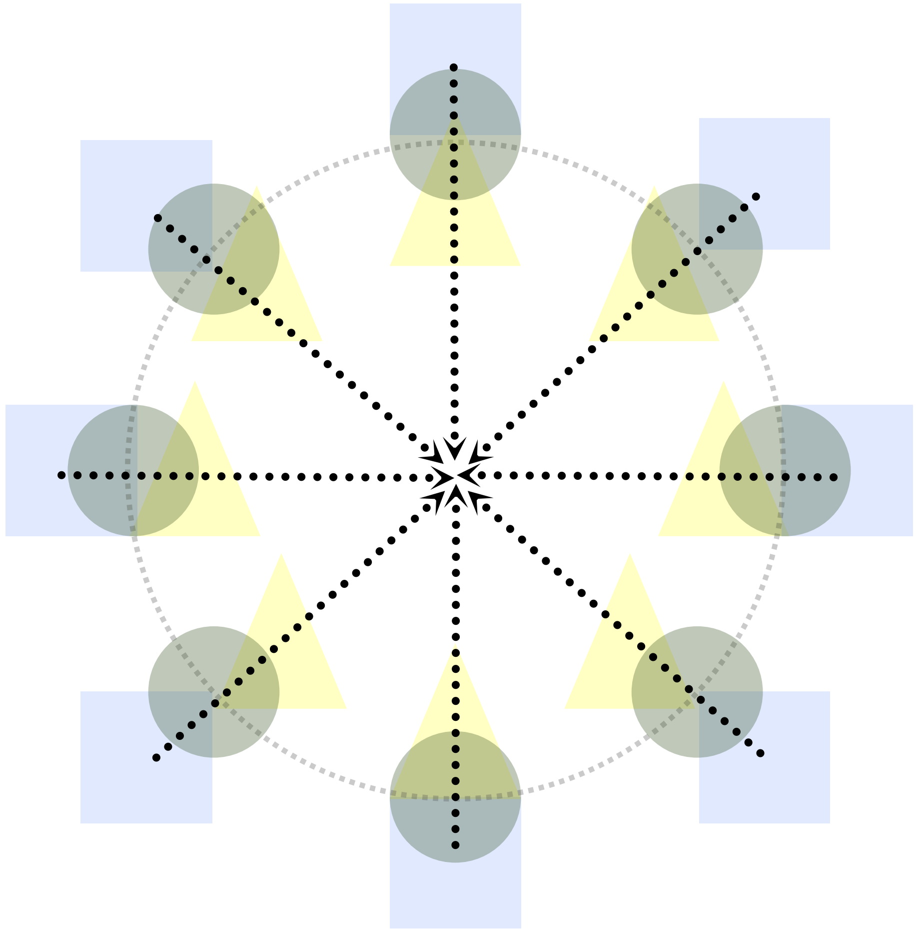

Step1: Digitally SHIFT image to the focal plane – Blue Square

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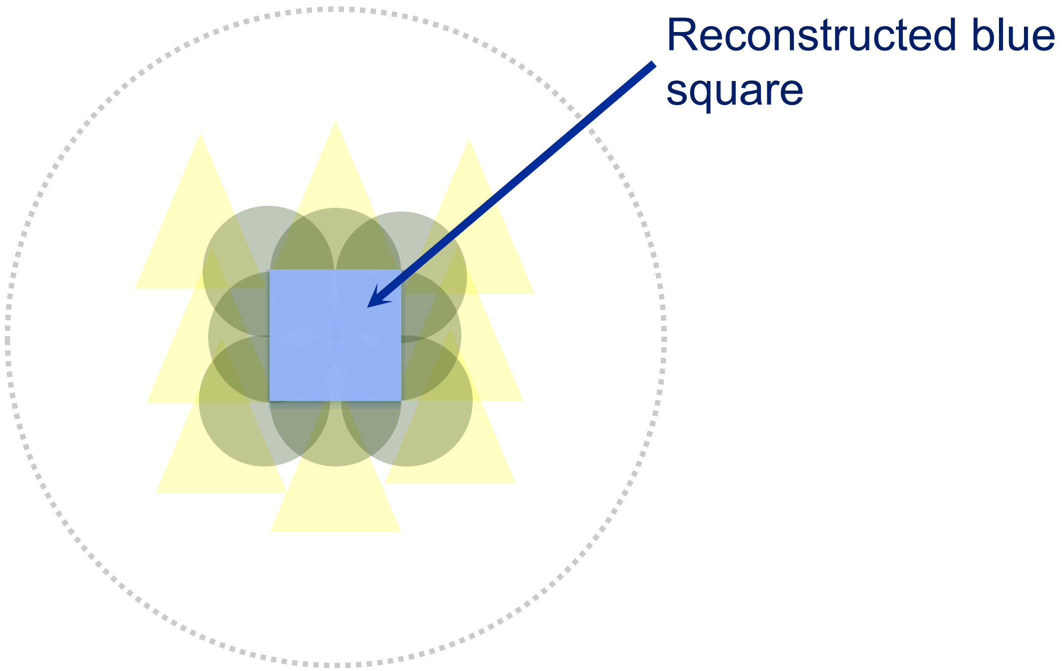
Image Reconstruction Process



Step2: Digitally ADD the SHIFTED image

Image Reconstruction Process

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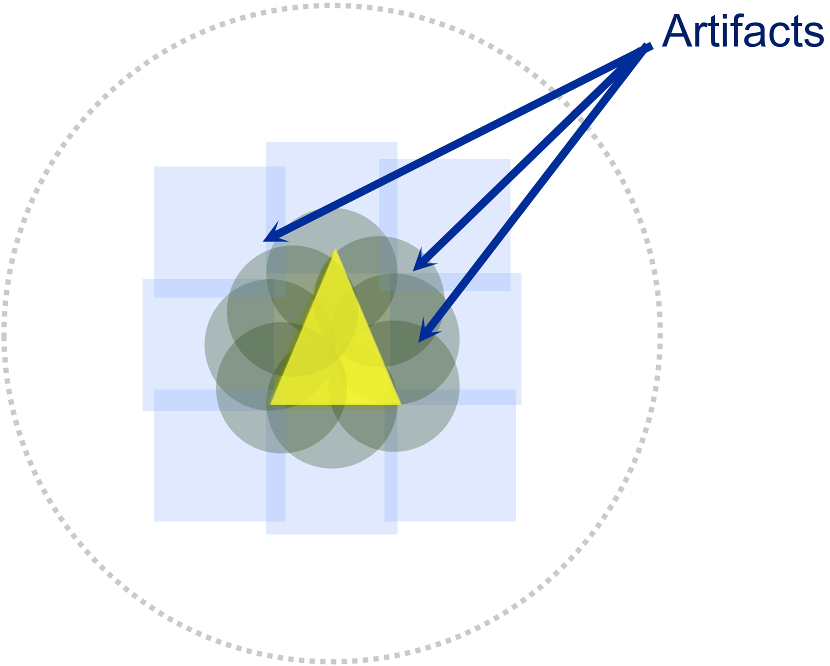


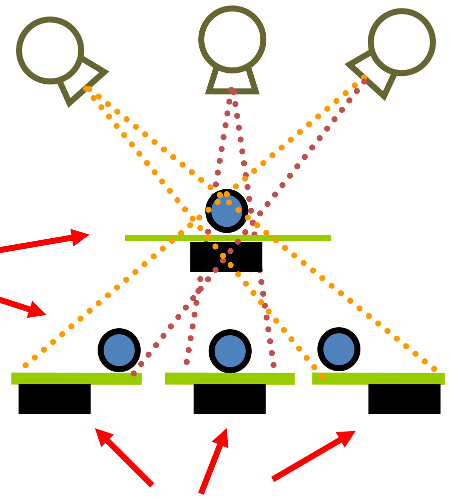
Step 3: Reconstruct Yellow Triangle

X-Ray Digital Tomosynthesis

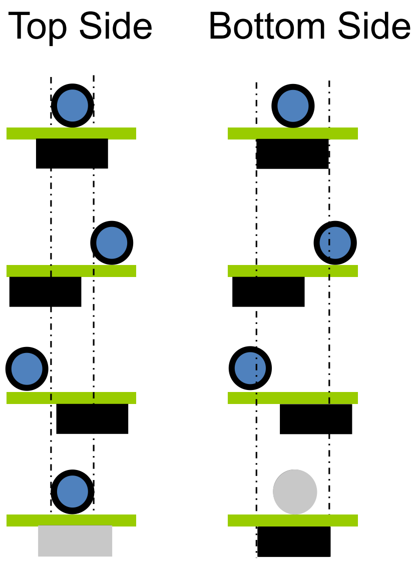
- Simple explanation of how it works:

Acquire multiple 2D projection images

Utilize parallax to separate Top side from Bottom side of PCB.



Need multiple images to separate all top side features from bottom side.



Step 1: Select image

Step 2: Align Image 2 and combine

Step 3: Align image 3 and combine

Resulting Combined Image: 1 + 2 + 3

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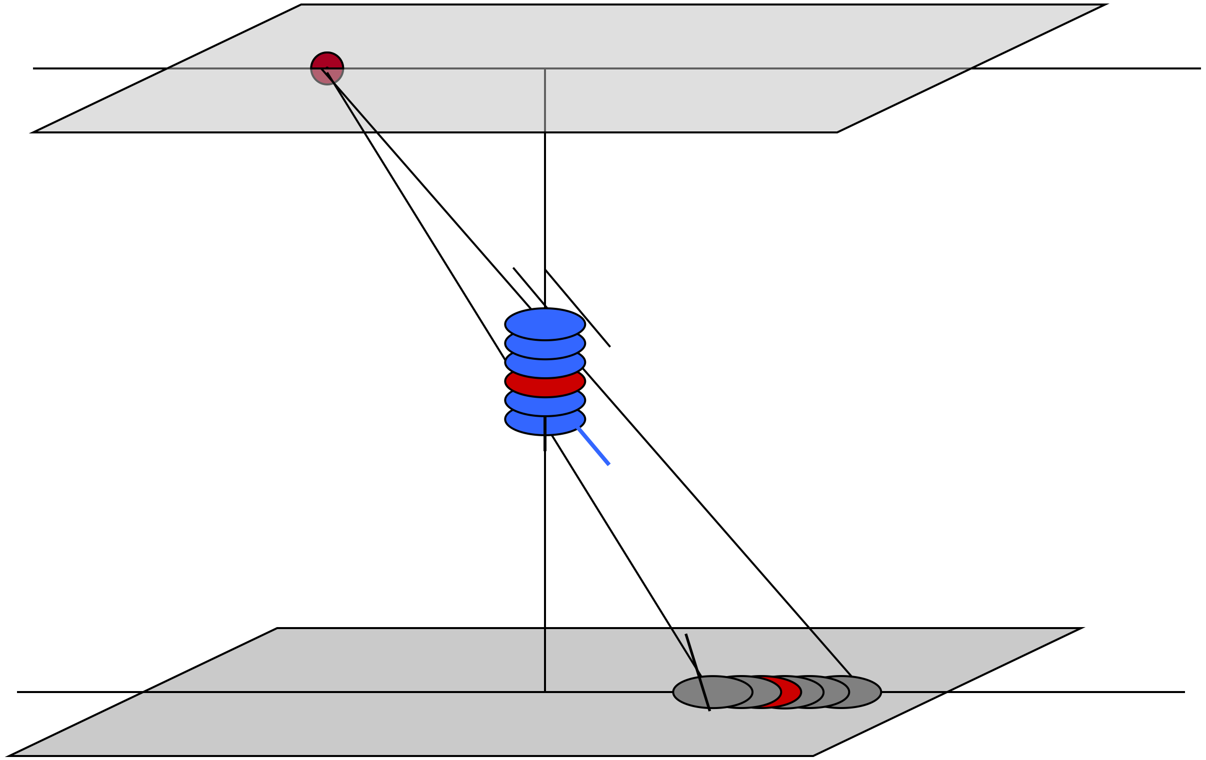
X-Ray Digital Tomosynthesis



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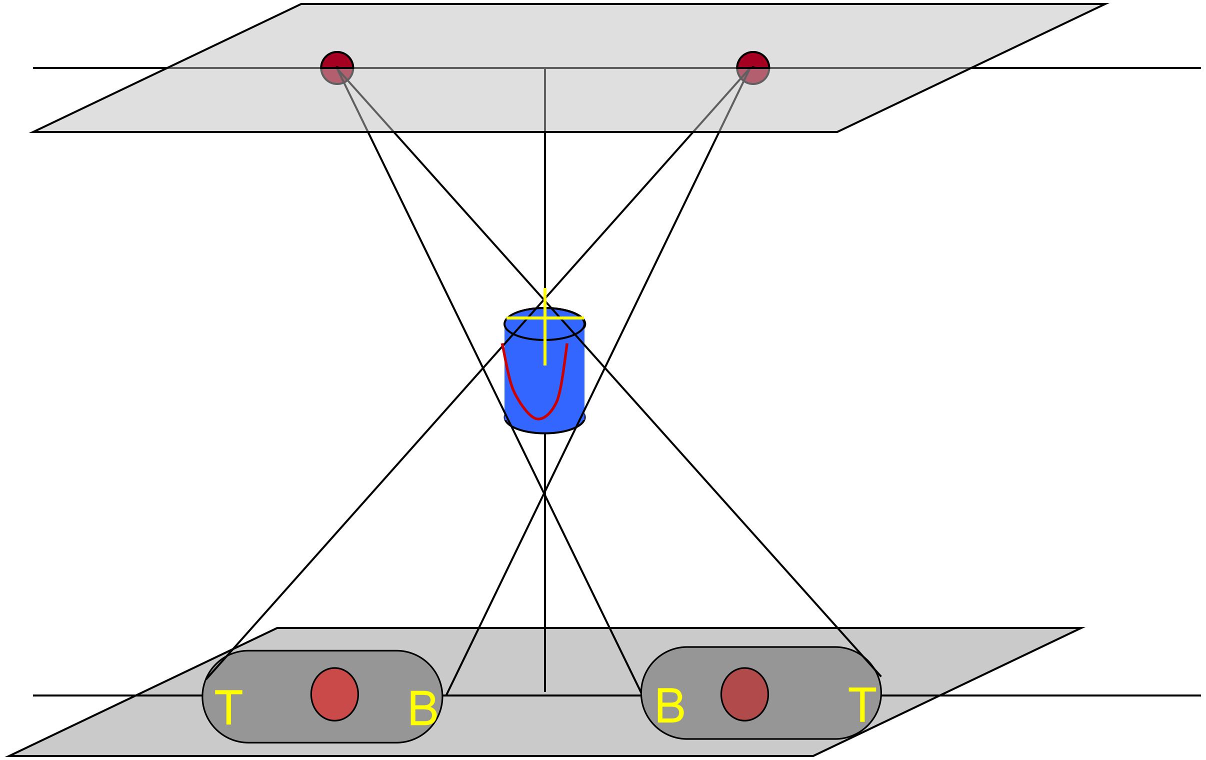
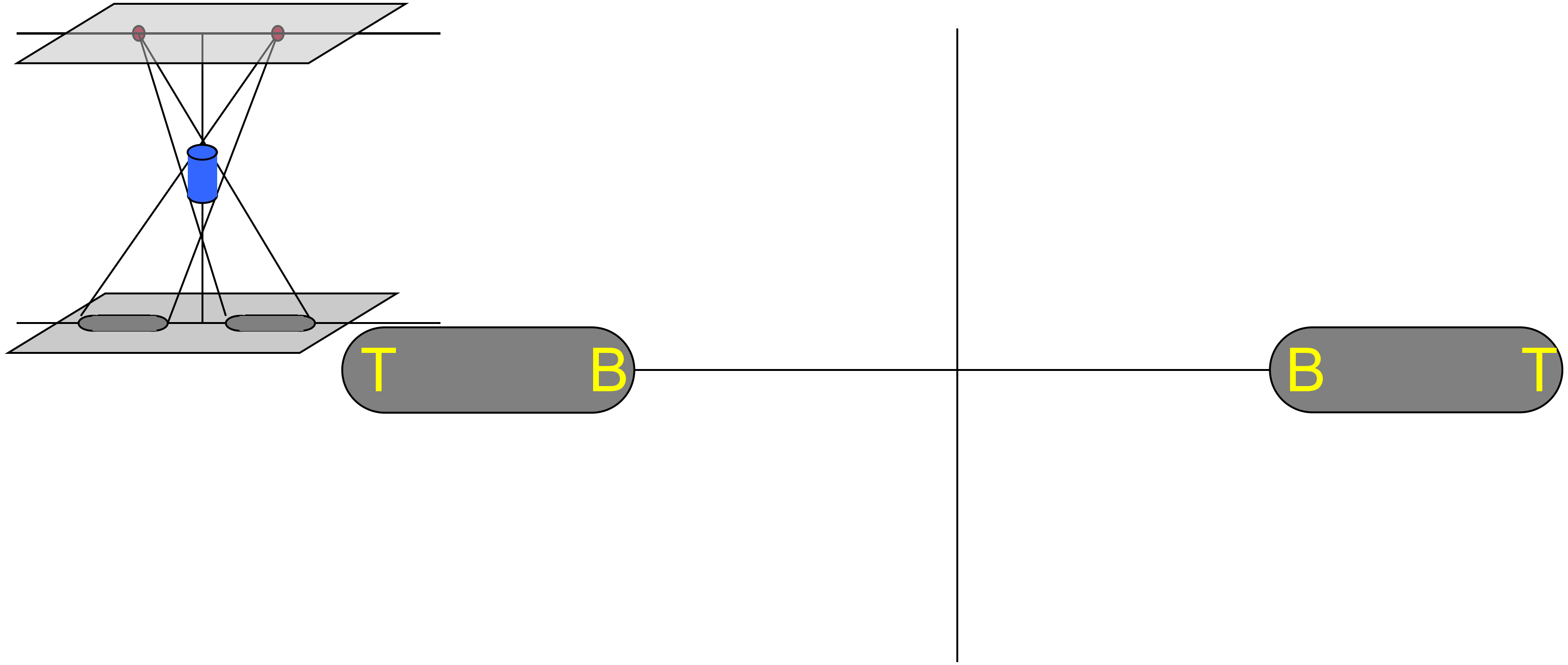
X-Ray Digital Tomosynthesis



X-Ray Digital Tomosynthesis



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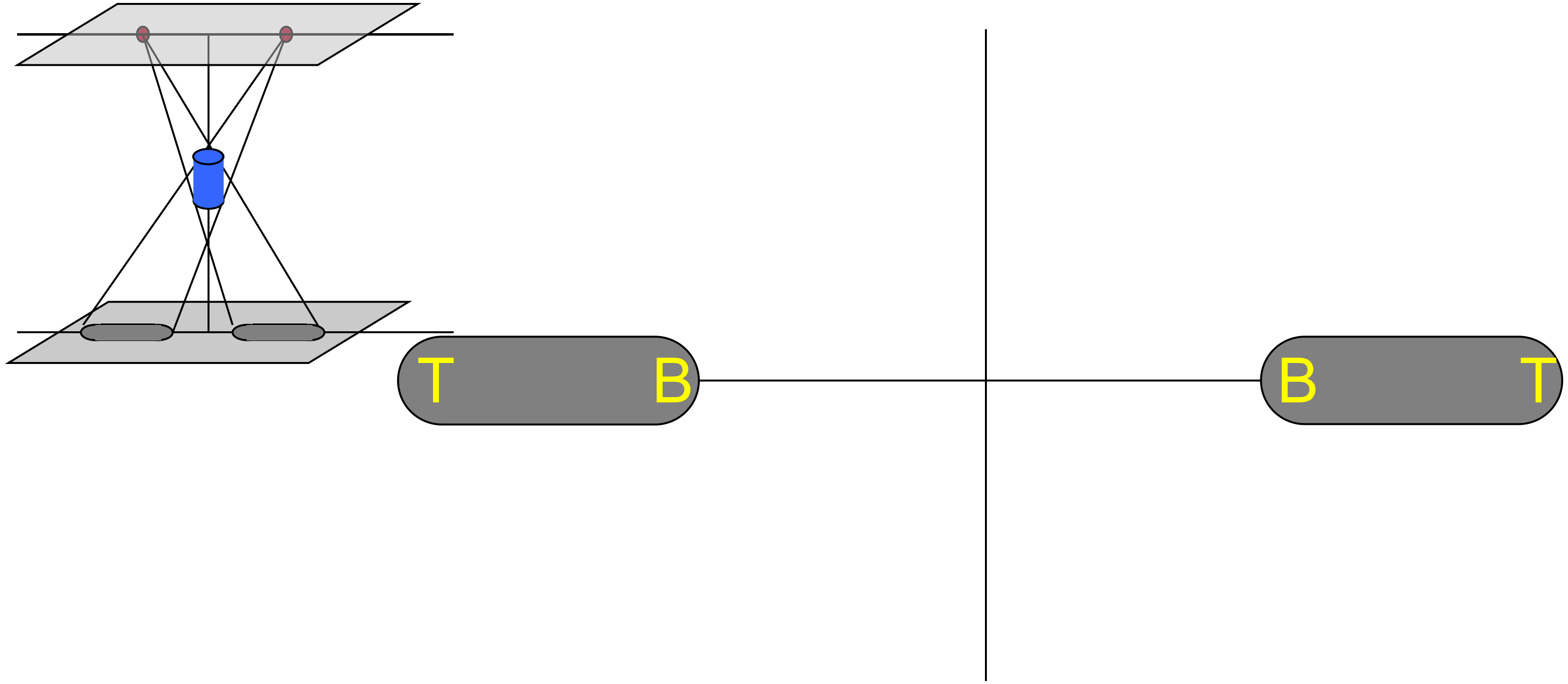
X-Ray Digital Tomosynthesis



X-Ray Digital Tomosynthesis

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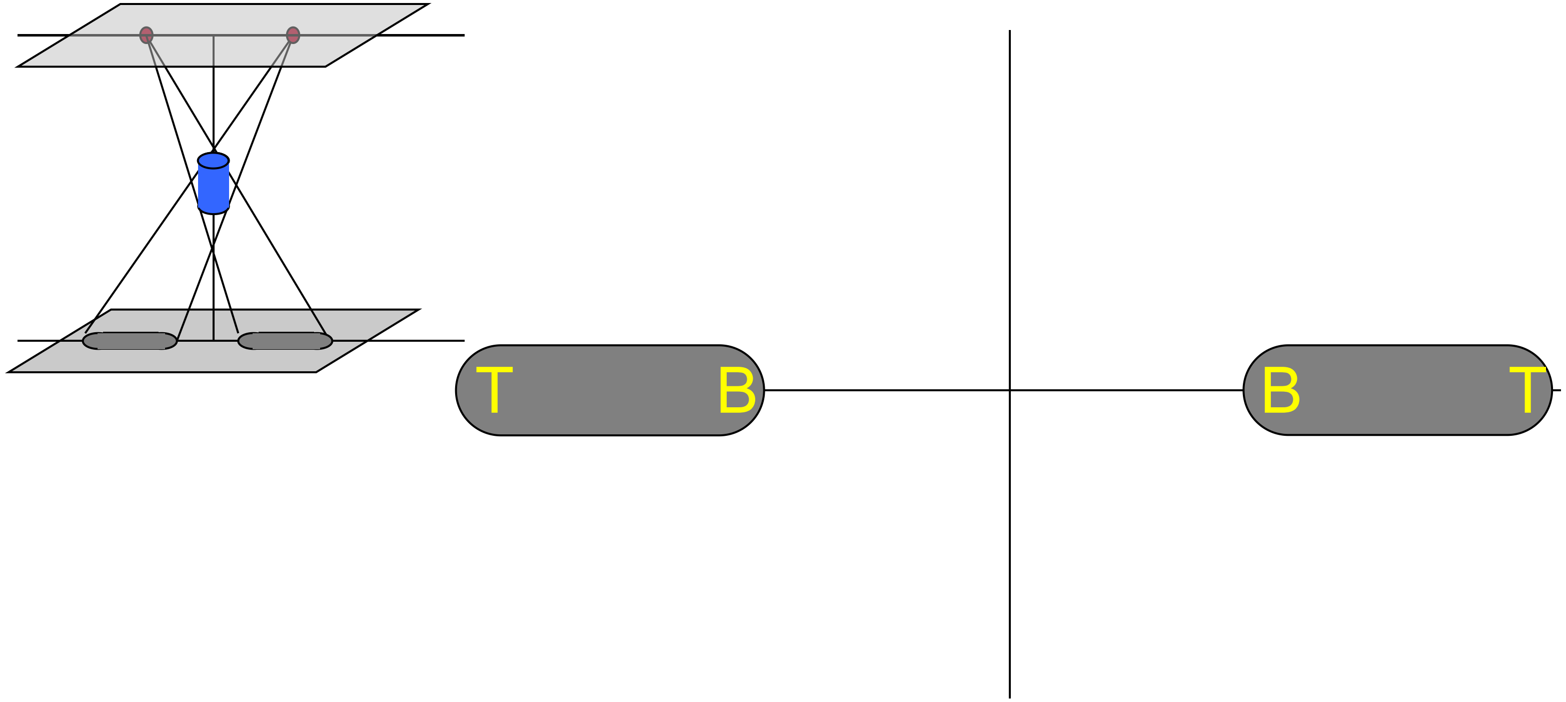

X-Ray Digital Tomosynthesis

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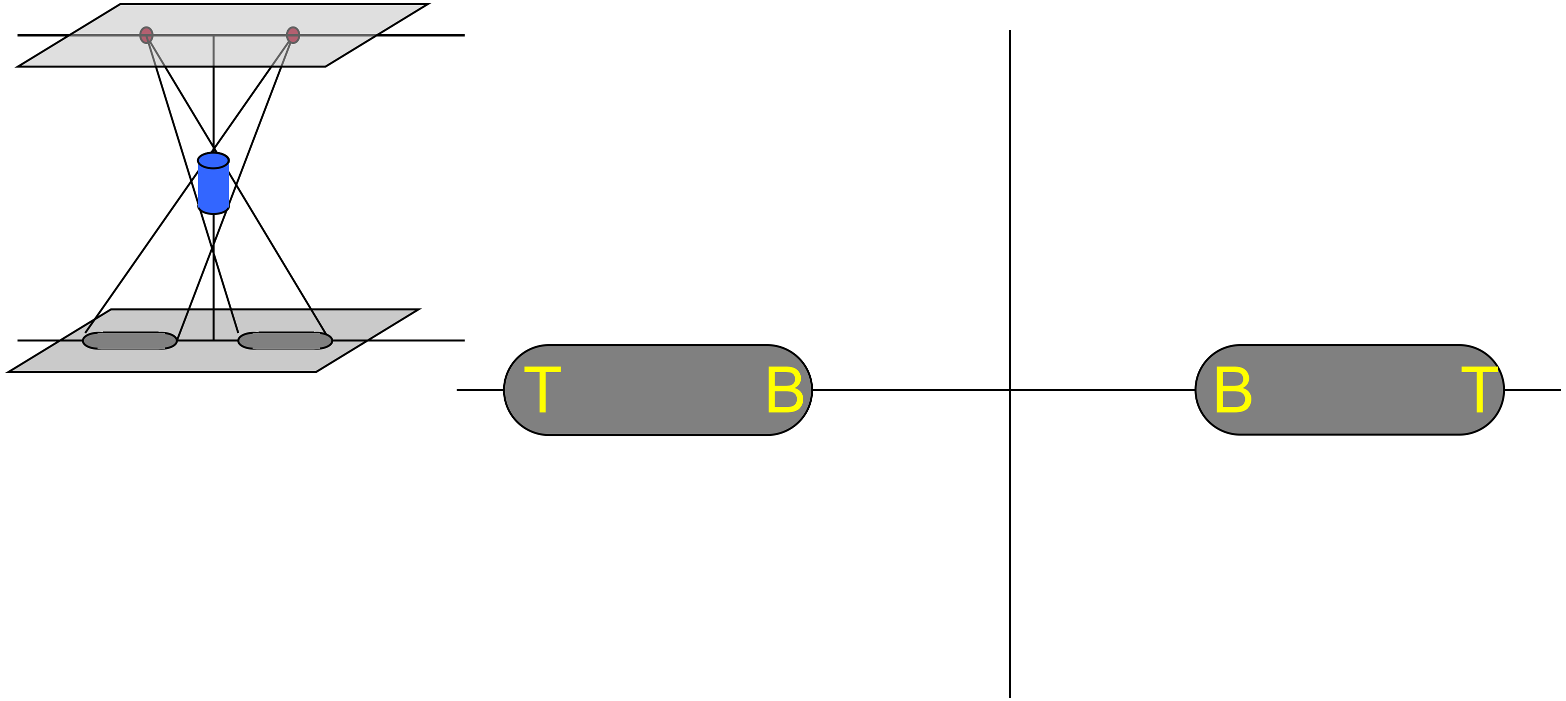
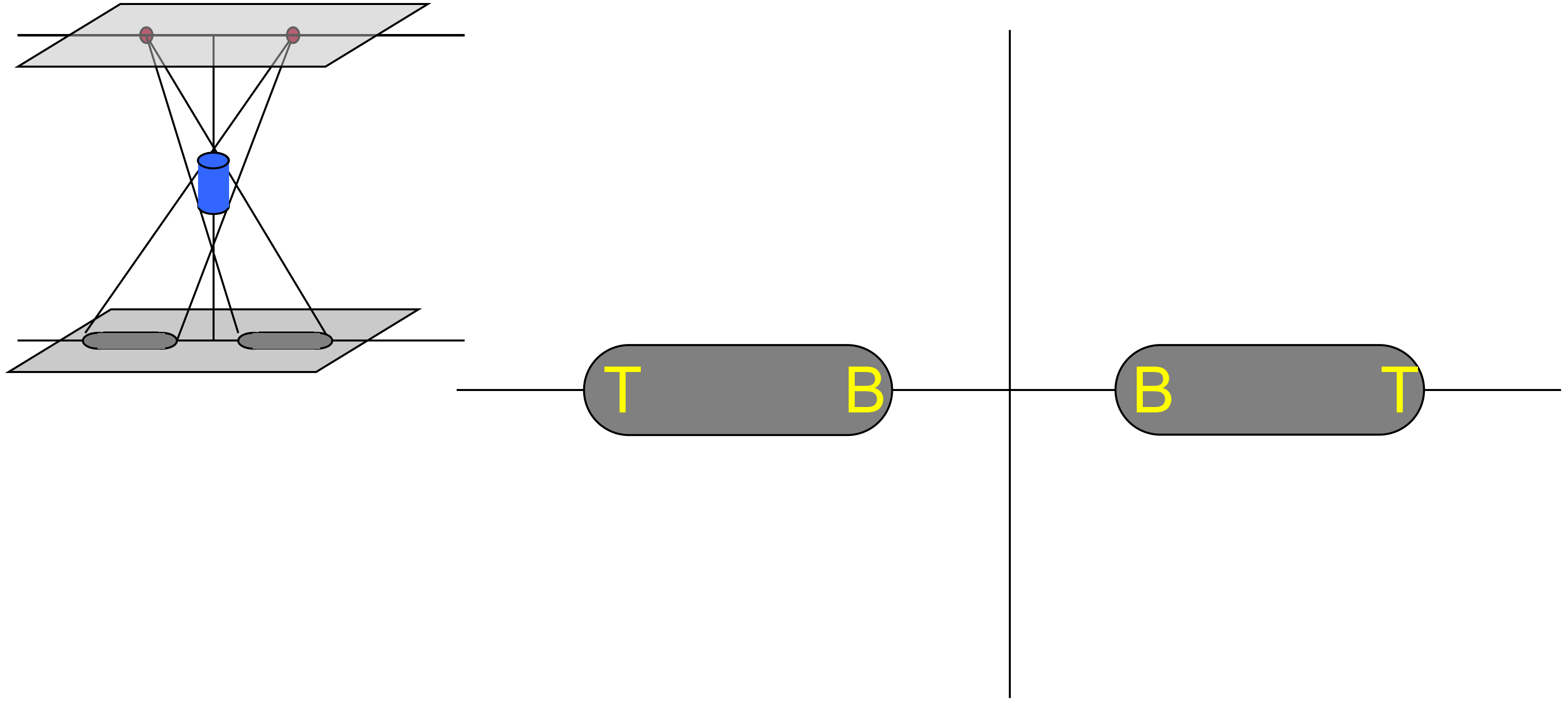


X-Ray Digital Tomosynthesis



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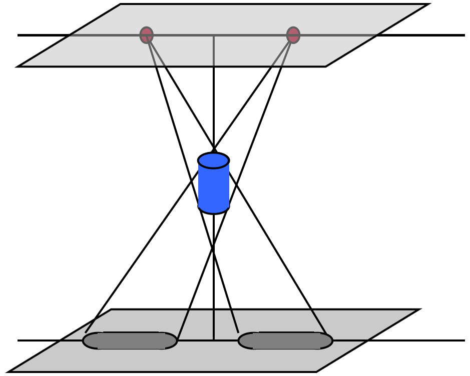
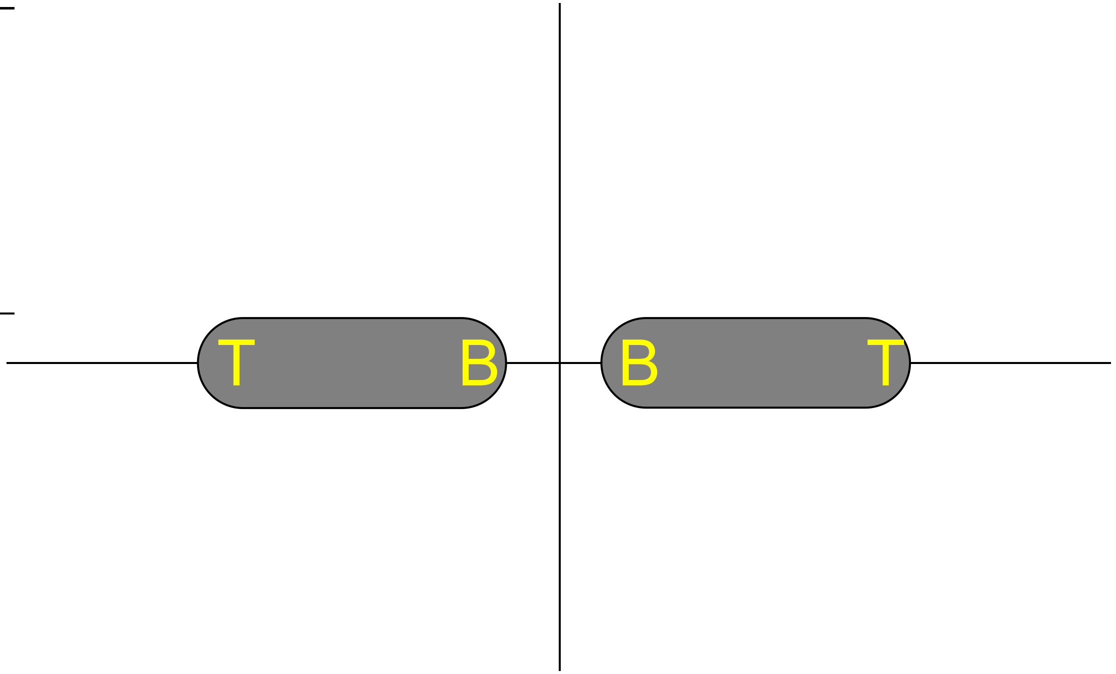
X-Ray Digital Tomosynthesis





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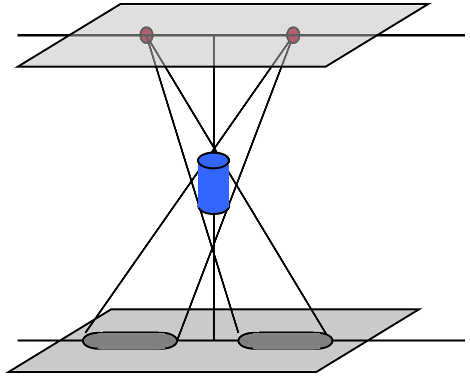
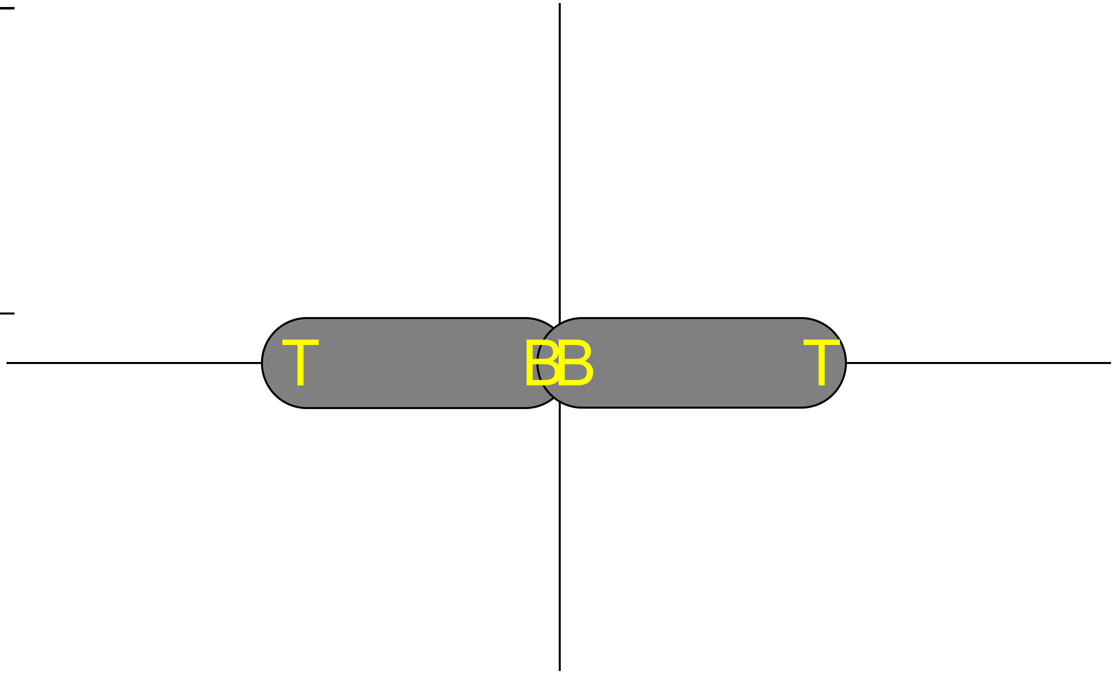
X-Ray Digital Tomosynthesis





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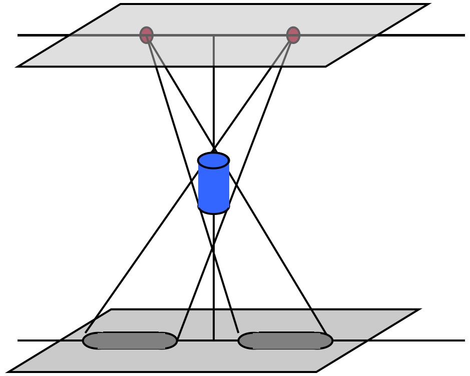
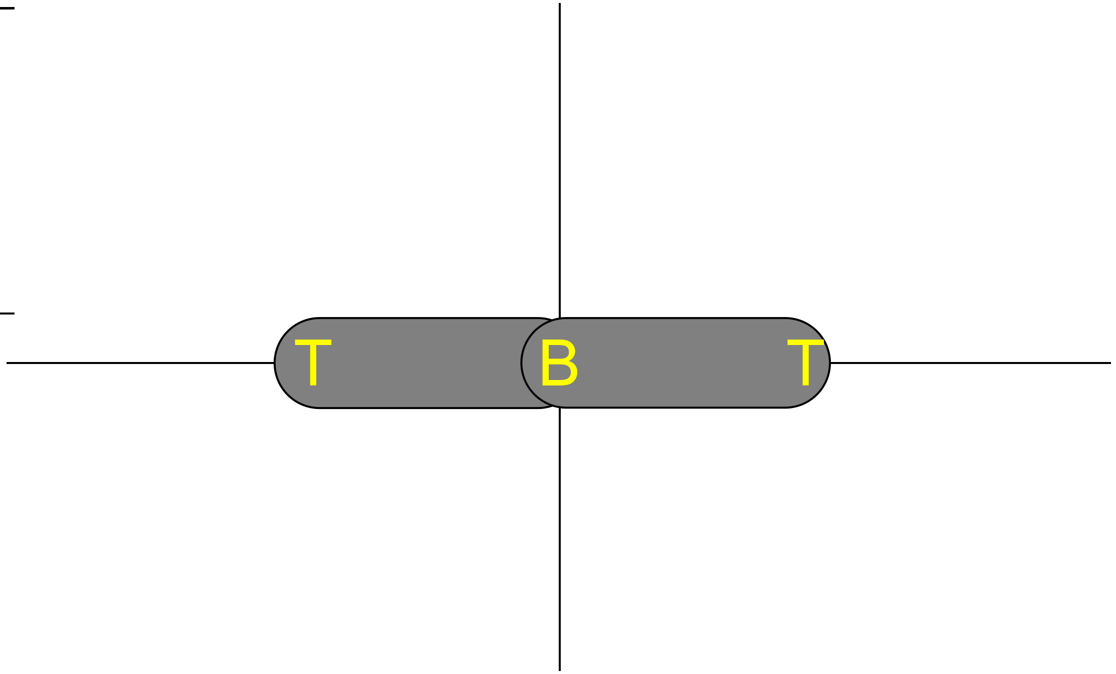
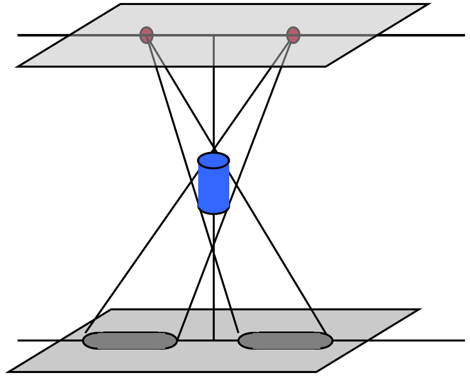
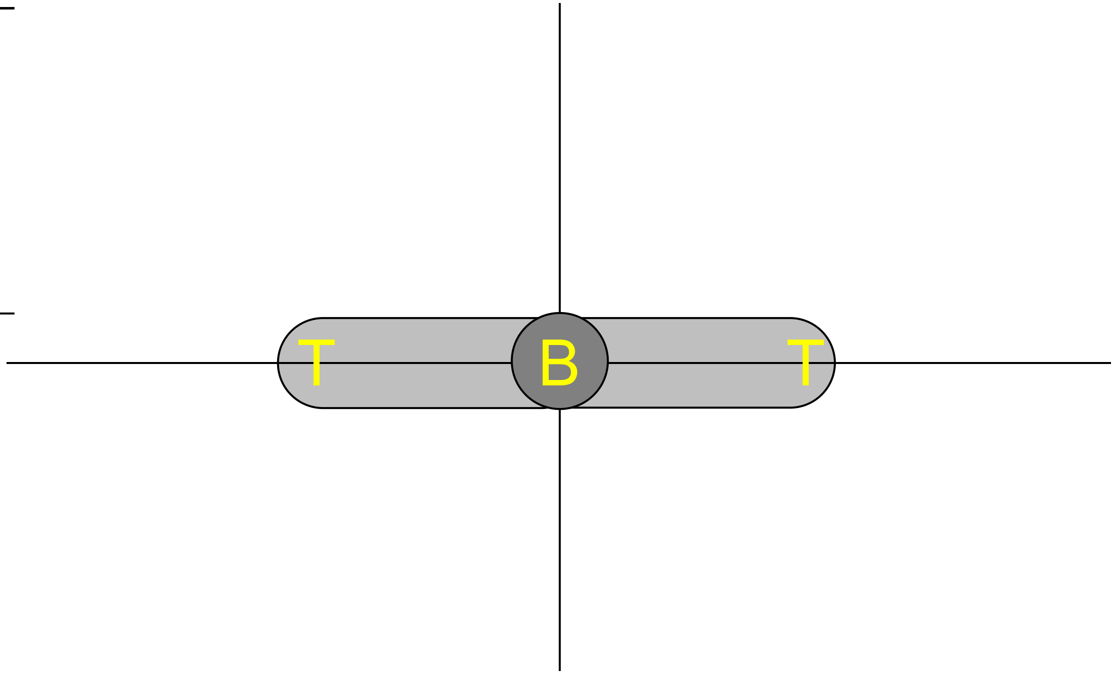
X-Ray Digital Tomosynthesis





X-Ray Digital Tomosynthesis





Shift and Add
1) Shift to Align
2) Add and divide (average) to emphasis the desired area

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Imaging Hardware Definition

X-Ray Tube - to generate X-Ray source where pass through scanning object to generate image (as emitter).

X-Ray Camera Array - As receiver to snap image projected X-Ray sources + scanning object. Acquires images and sends to IRPs via Ethernet cables and the Gigabit Switch

Camera Controller (Trigger board) – to provide Power and Triggers to Camera Array via Fire wire connections.

Gigabit Switch – Controls all image information between the Camera Array, the IRPs, and the System Controller

Image Reconstruction Processors – Reconstruct Digital images from the Camera Array

Y Axis Motion – Provides signals to Camera Controller to call cameras capture image

Digital I/O Assembly - Reads status of Camera Controller and controls Synthetic Triggers

Image Reconstruct Processor Driver

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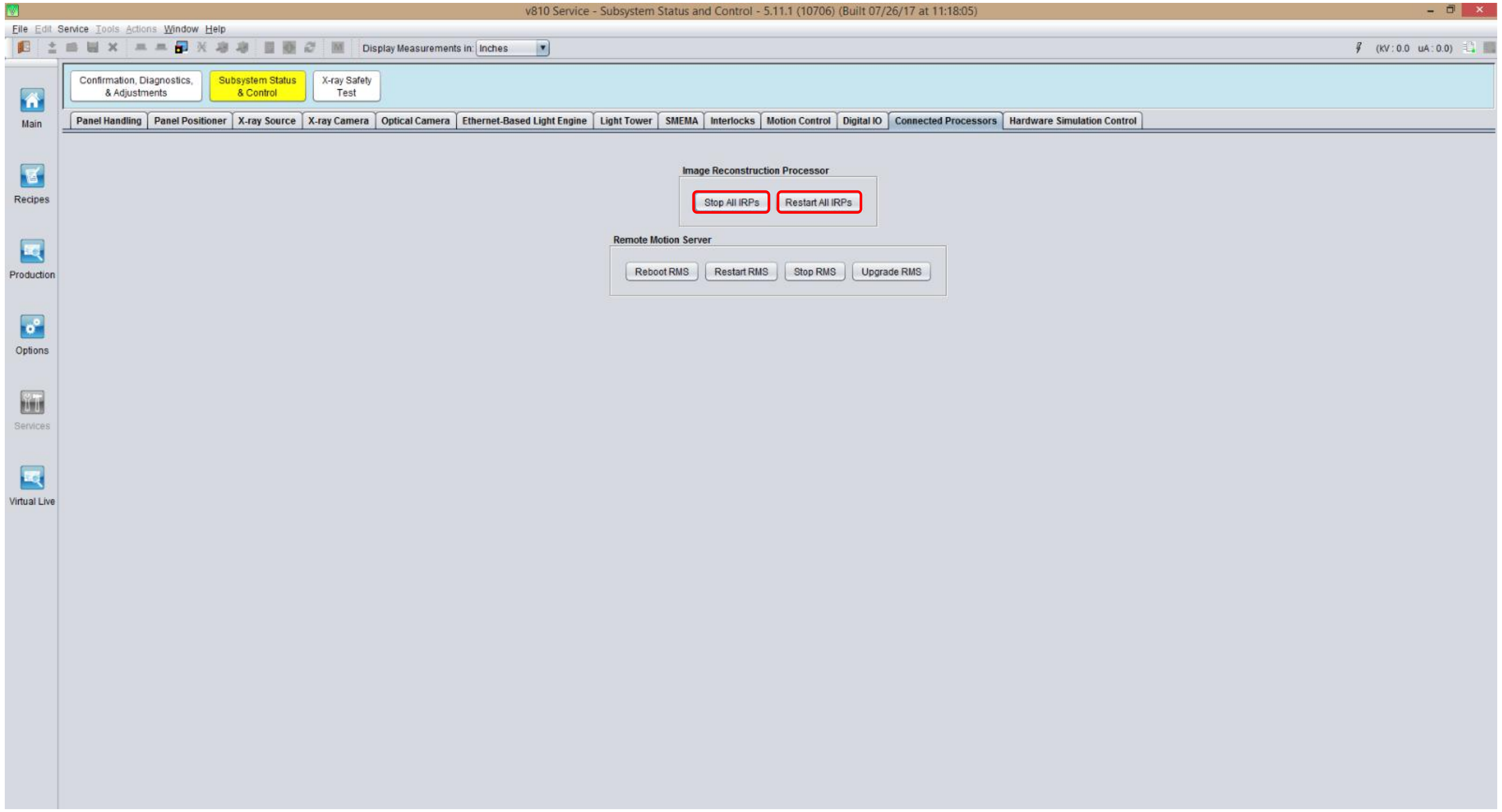


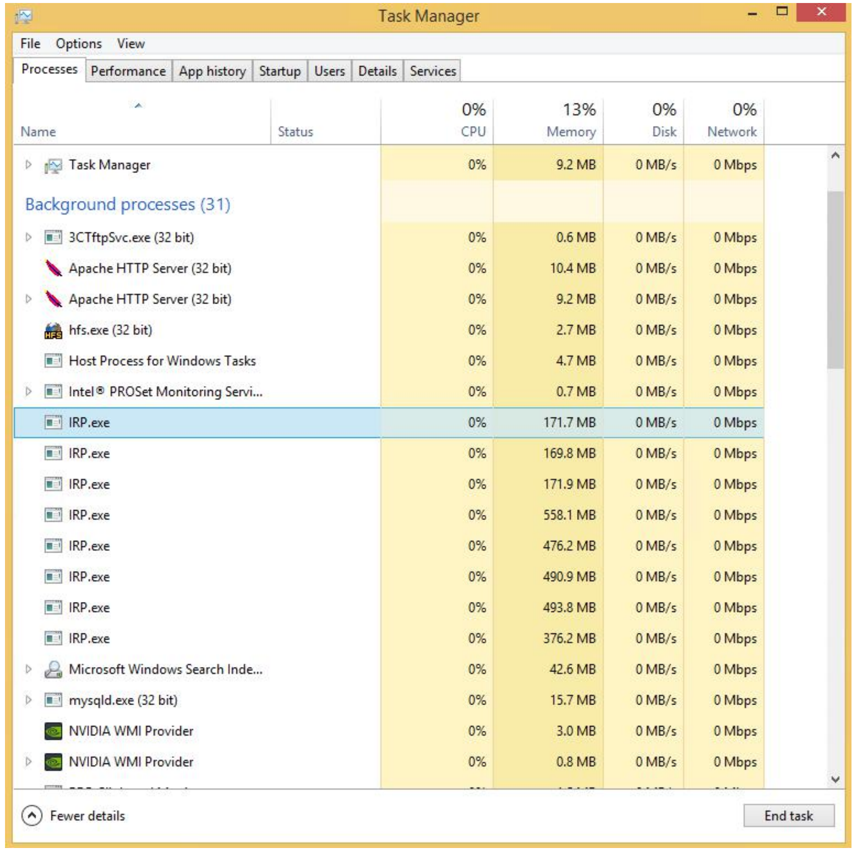
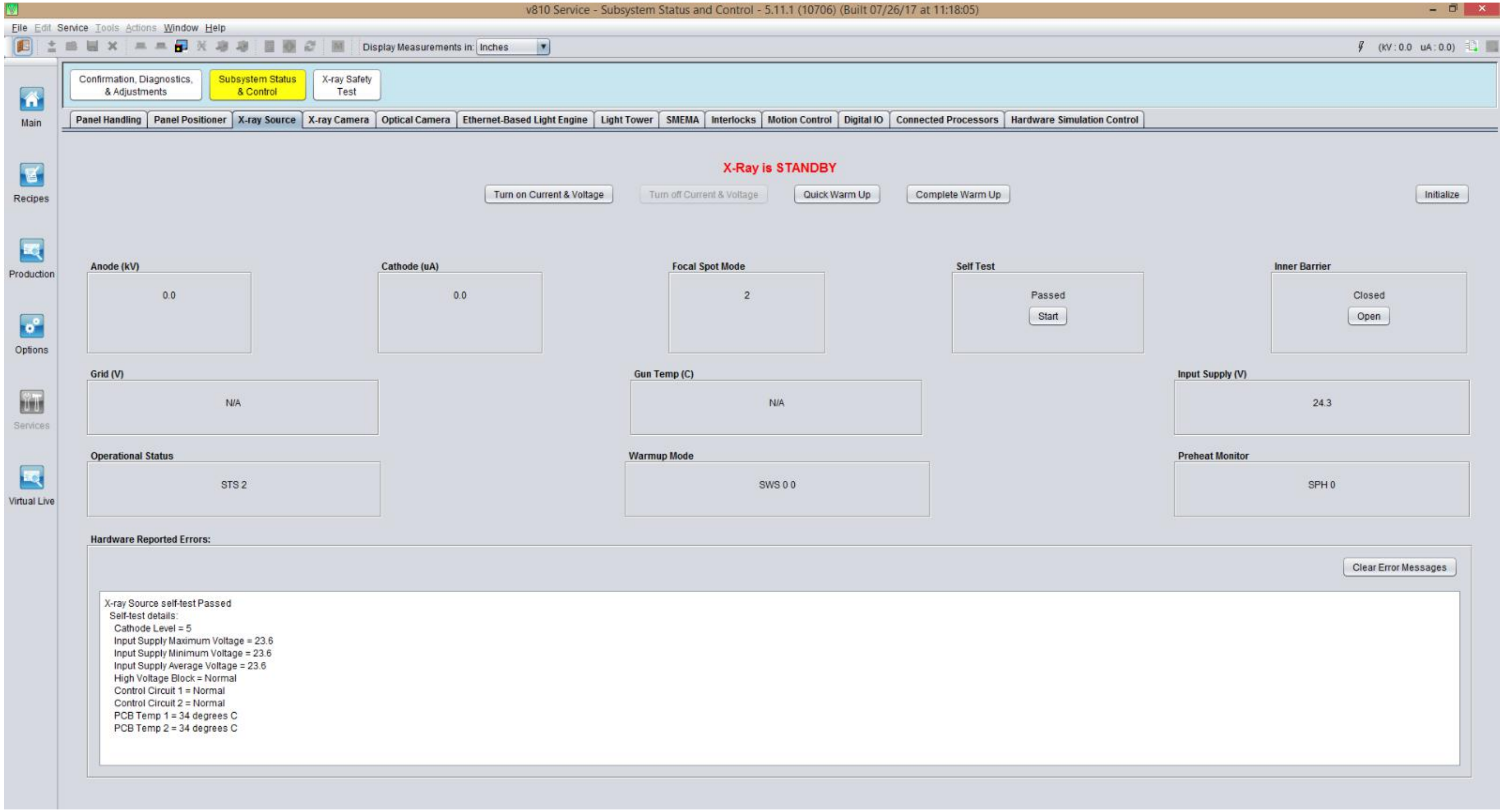


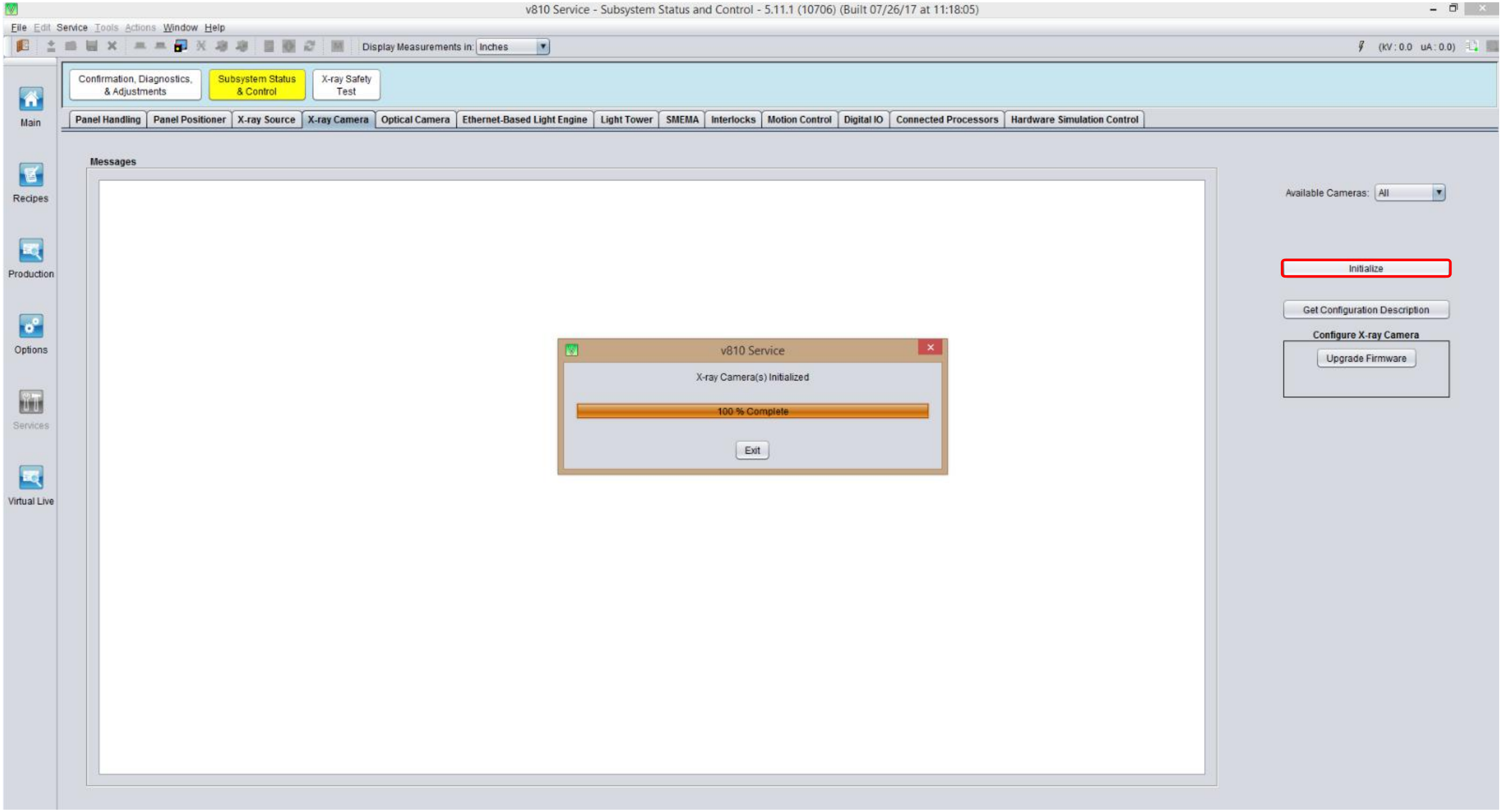
Image Reconstruct Processor Driver -
Installed at Main controller and main function to reconstruct image capture by 14 cameras 2D to 3D

X-Ray Source Tab



- Manual power On/Off X-Ray tube power
- Checking tube healthy condition

X-Ray Camera Tab



Hardware Overview - X-Ray Sources

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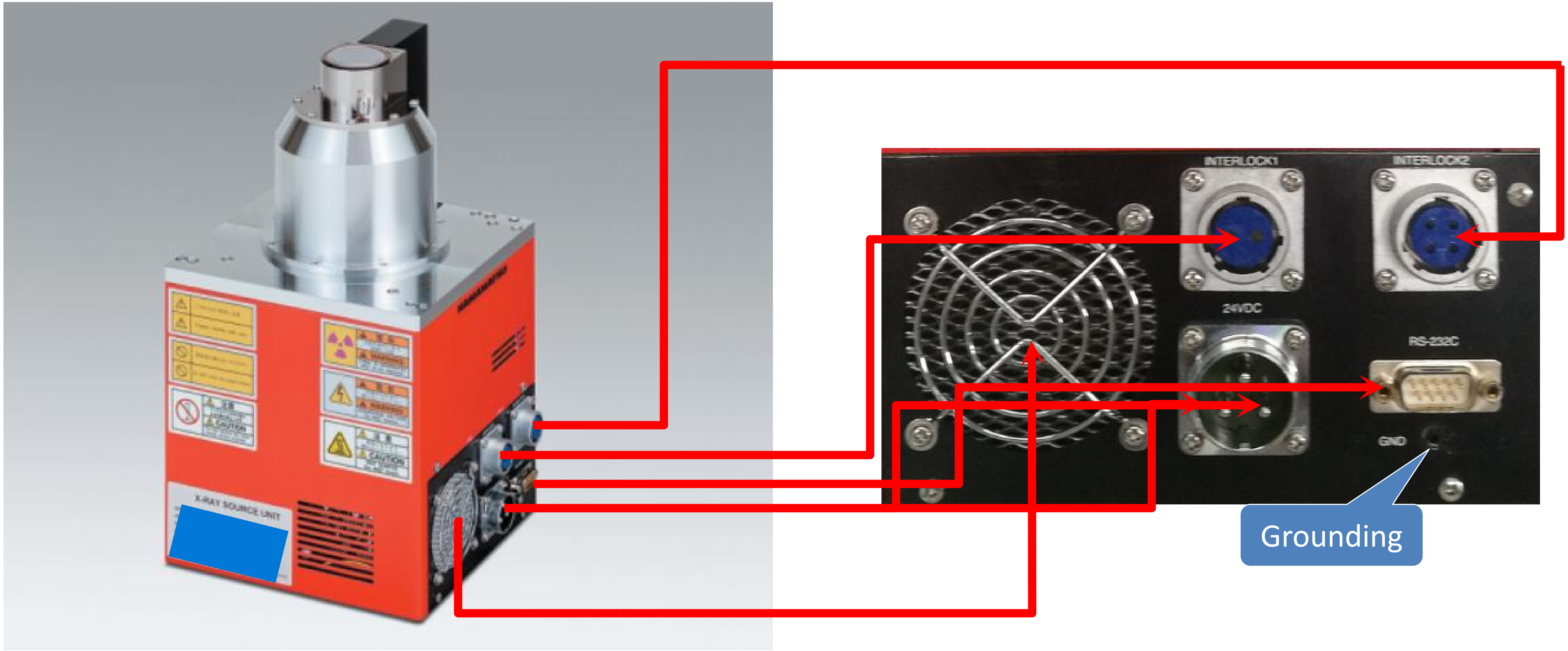

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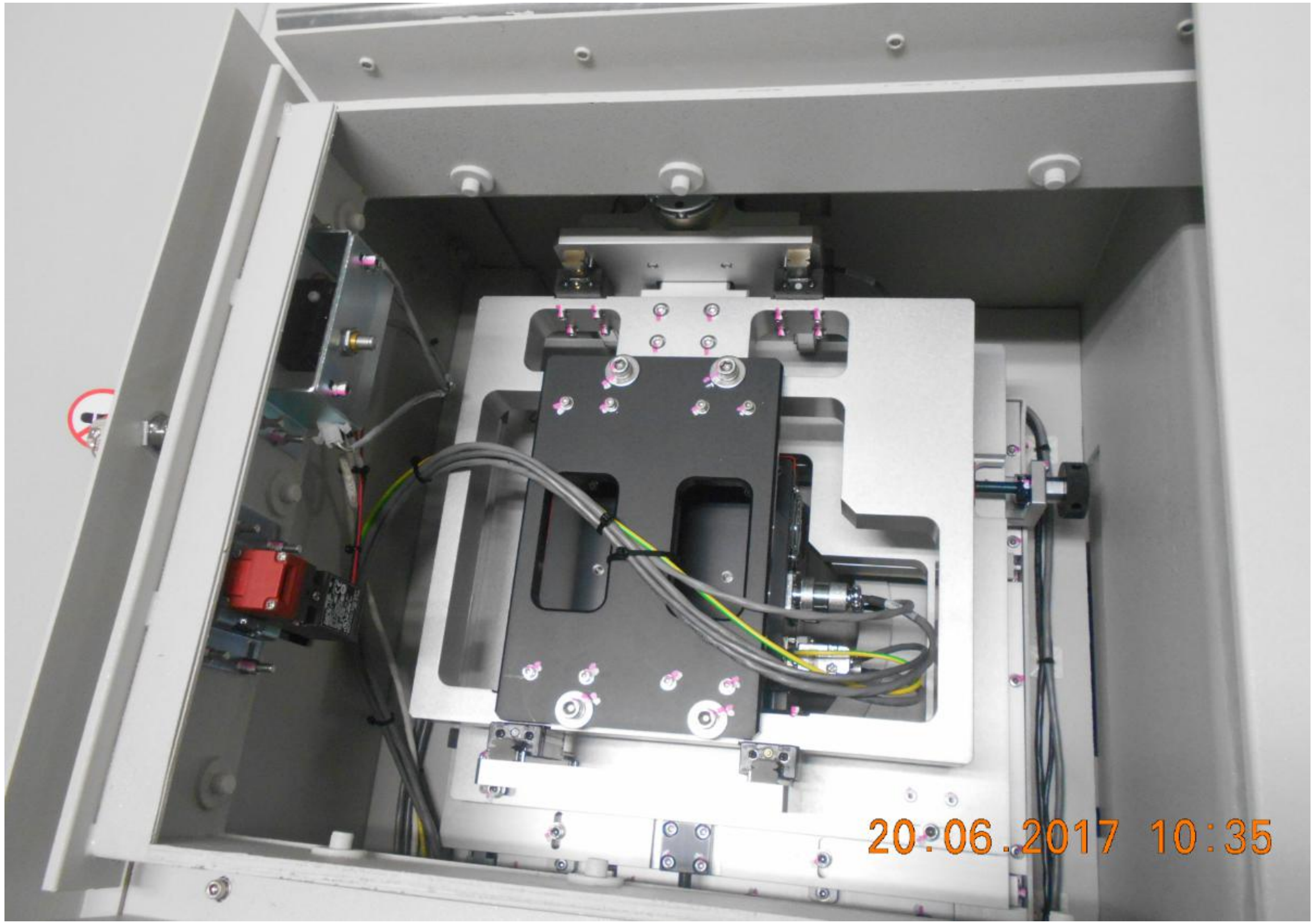
X-Ray Tube Placement



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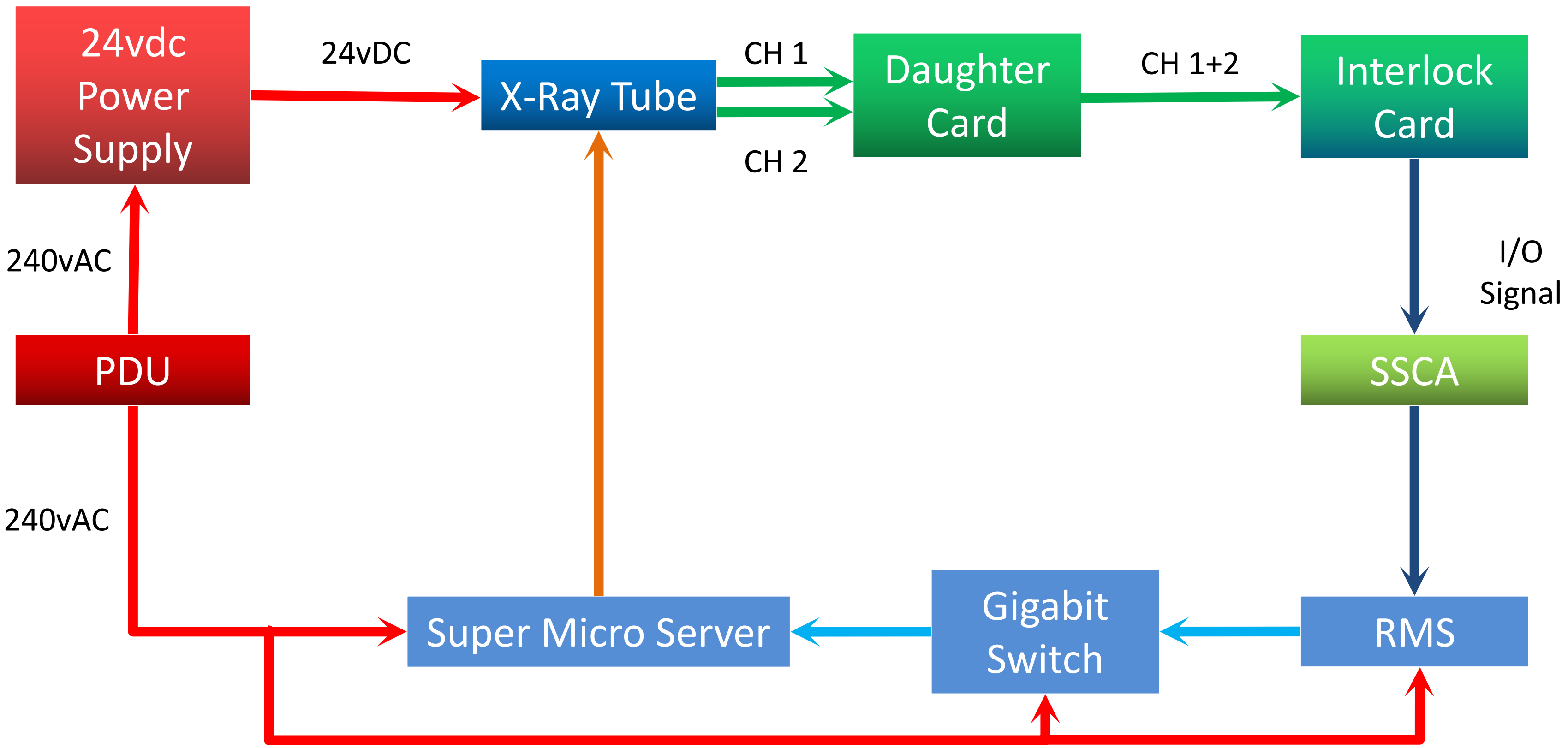
Xray Sources Connection Diagram



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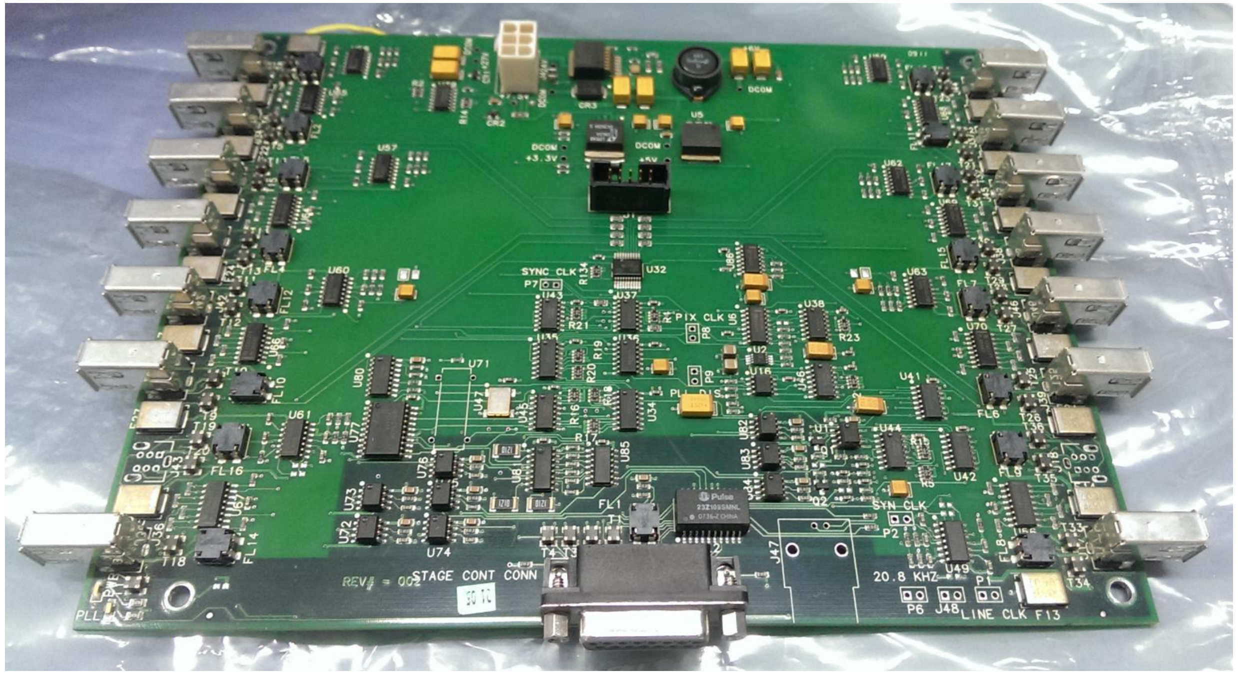
Hardware Overview - Trigger Board



Top

Bottom

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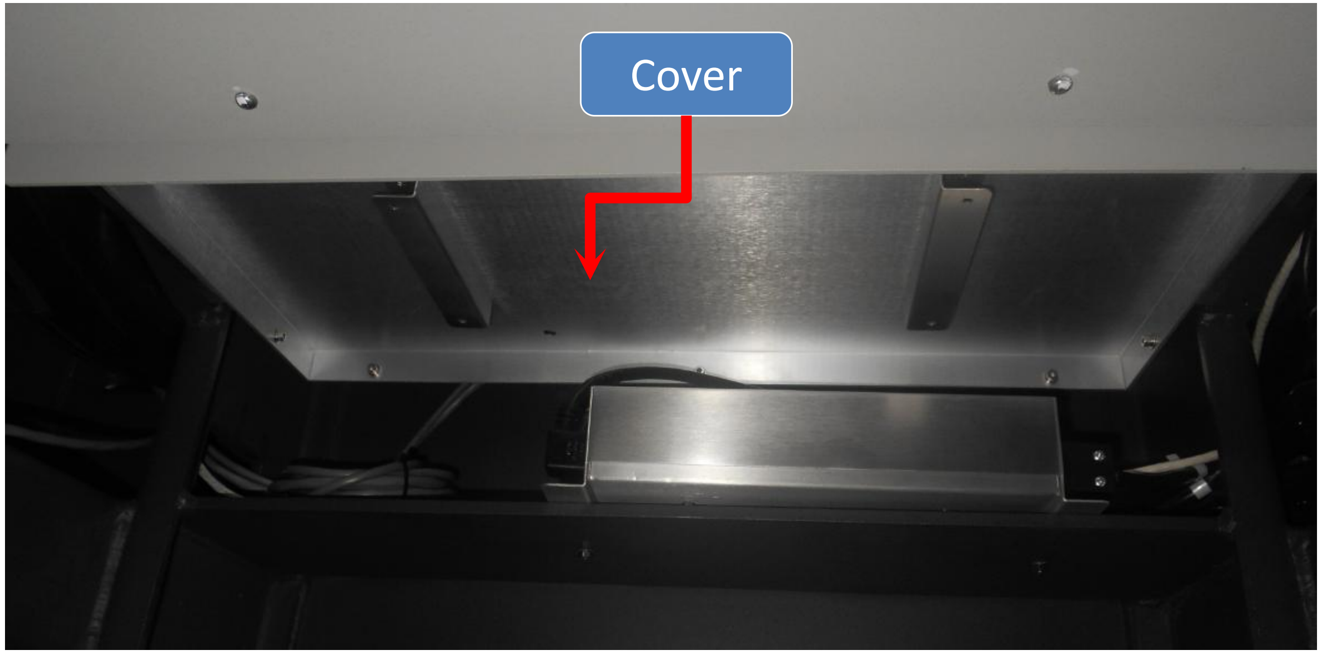
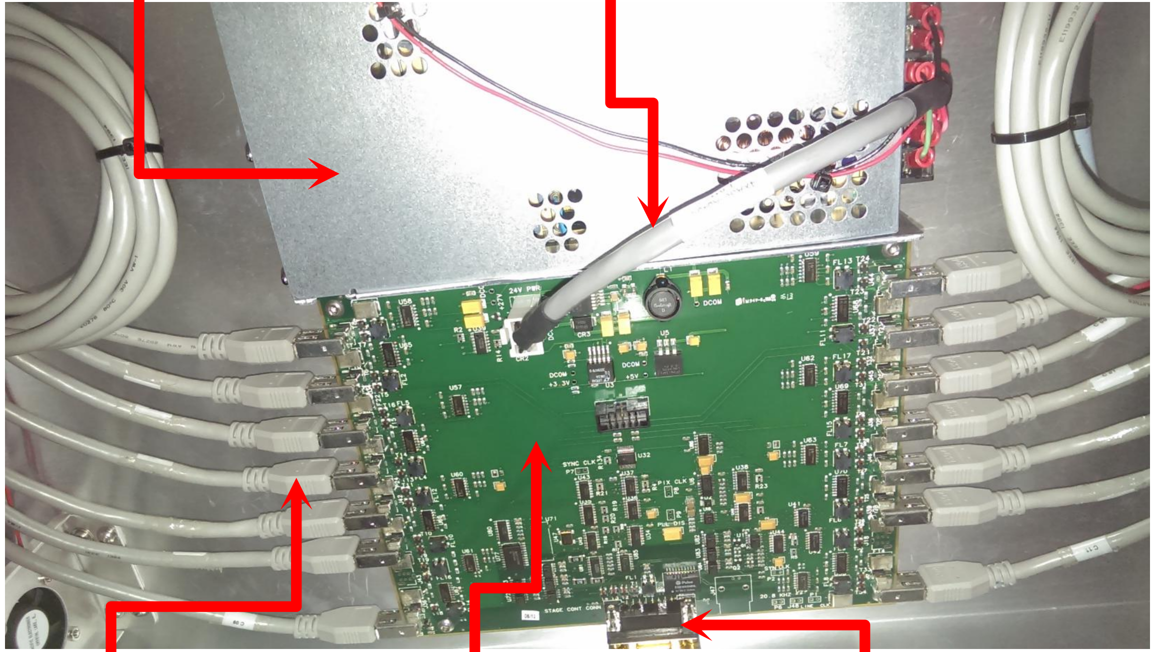
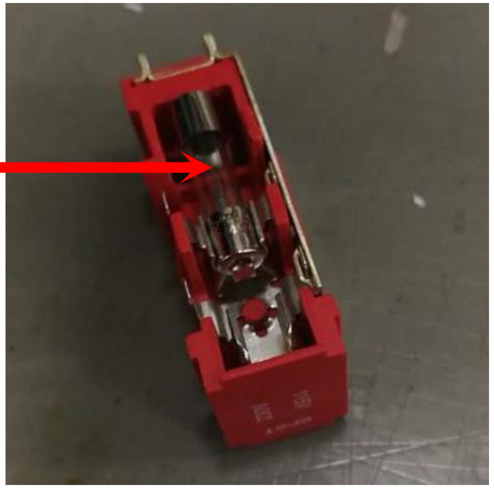
Hardware Overview - Trigger Board





Hardware Overview - Trigger Board



Fuse, 250vAC , 3A

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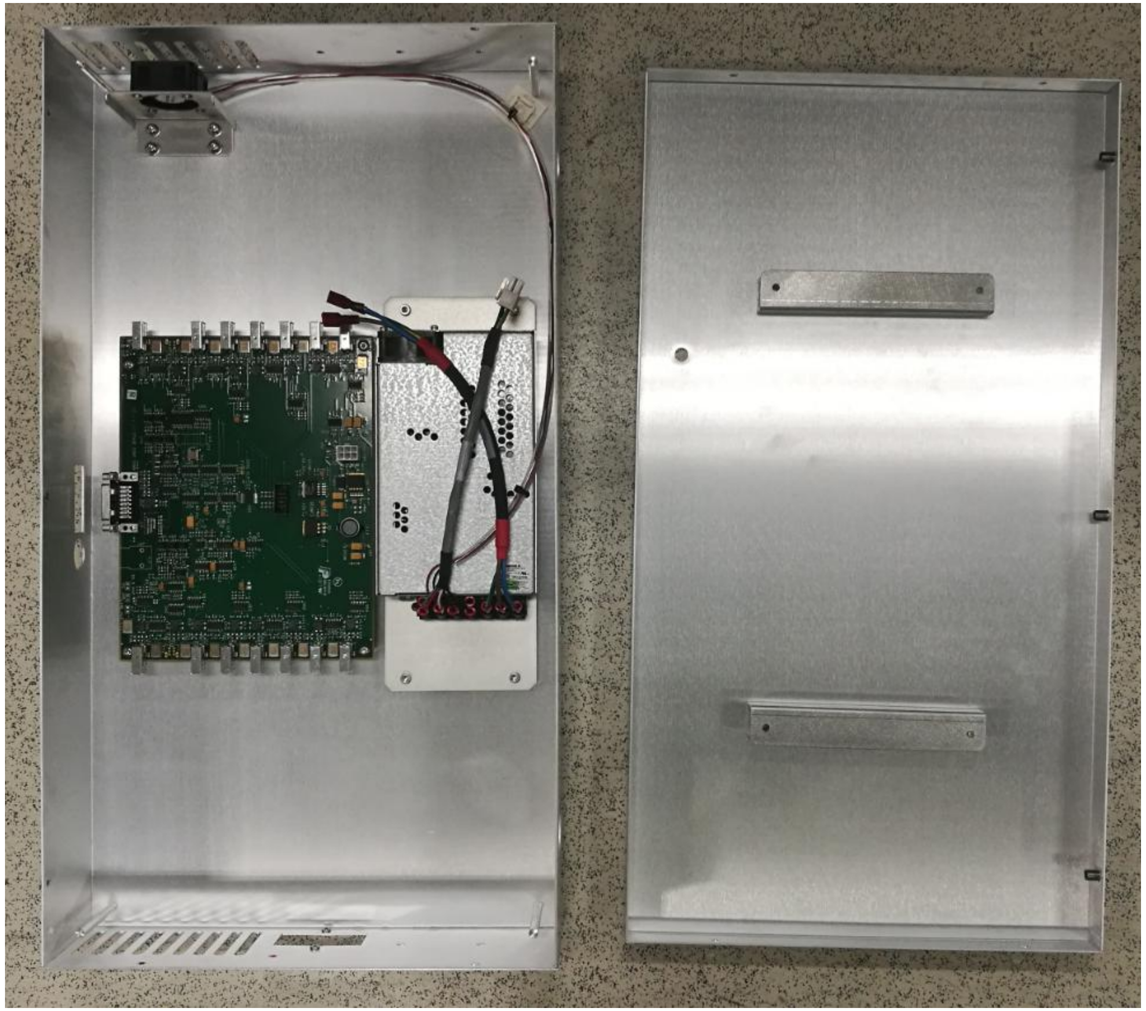
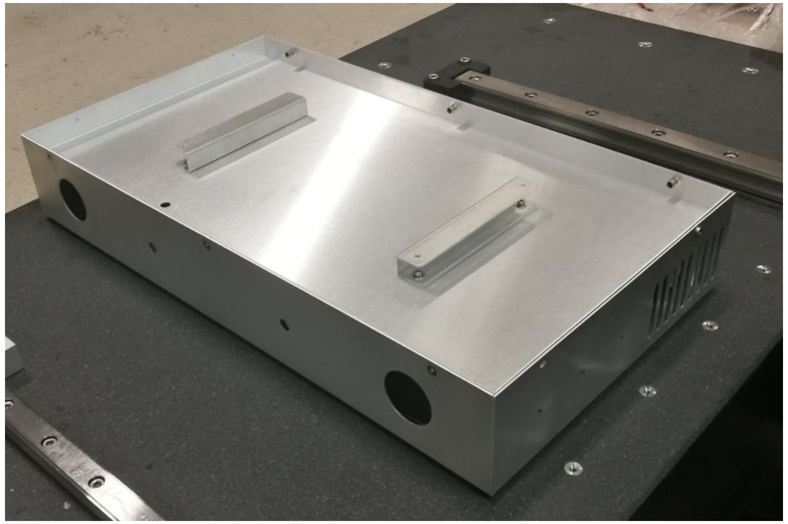
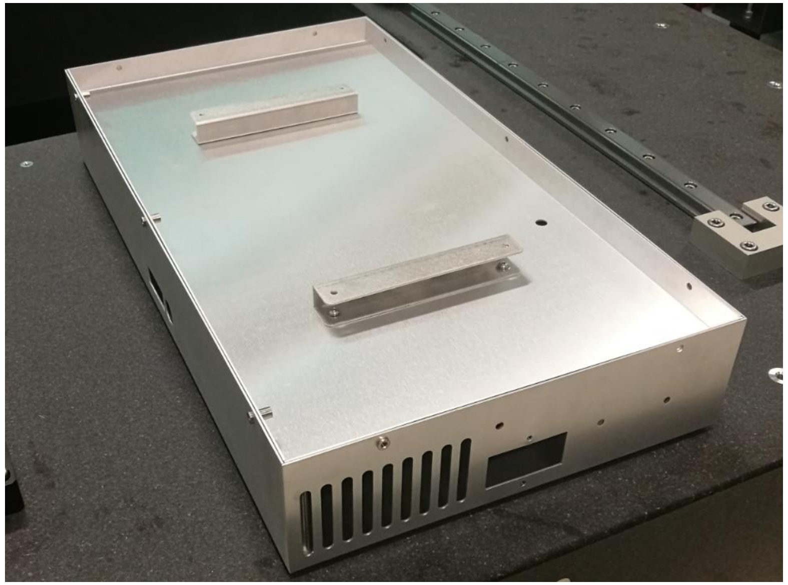
Hardware Overview - Trigger Board







Hardware Overview - X-Ray Camera- Gen 1

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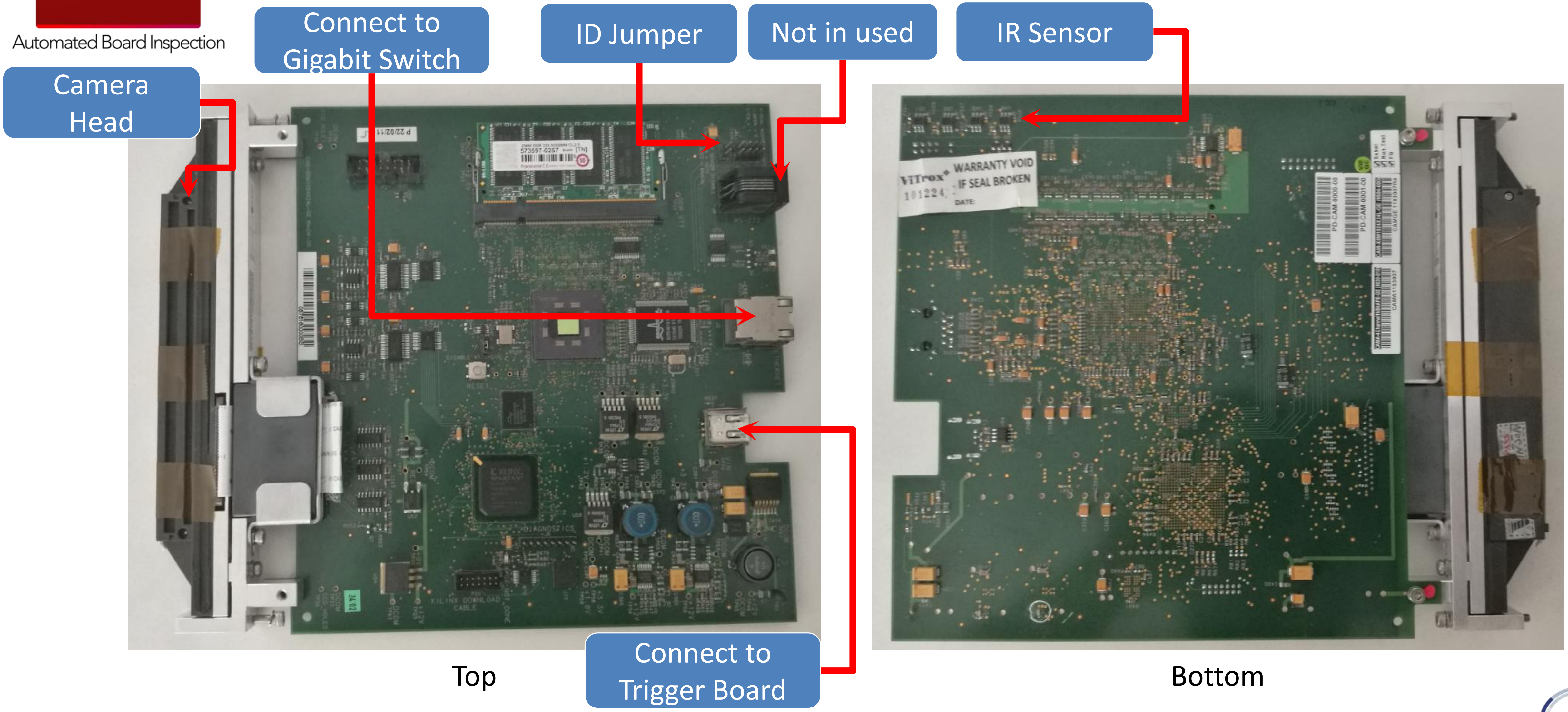

Top

Bottom

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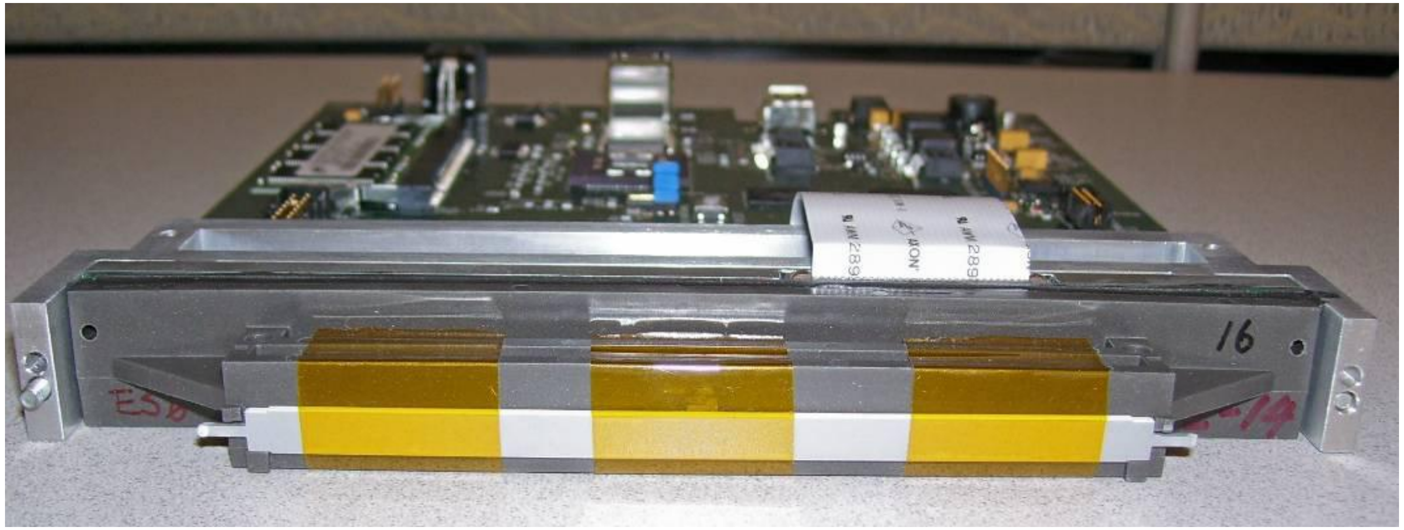
Hardware Overview - X-Ray Camera- Gen 1



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Hardware Overview - X-Ray Camera- Gen 3 Single Port

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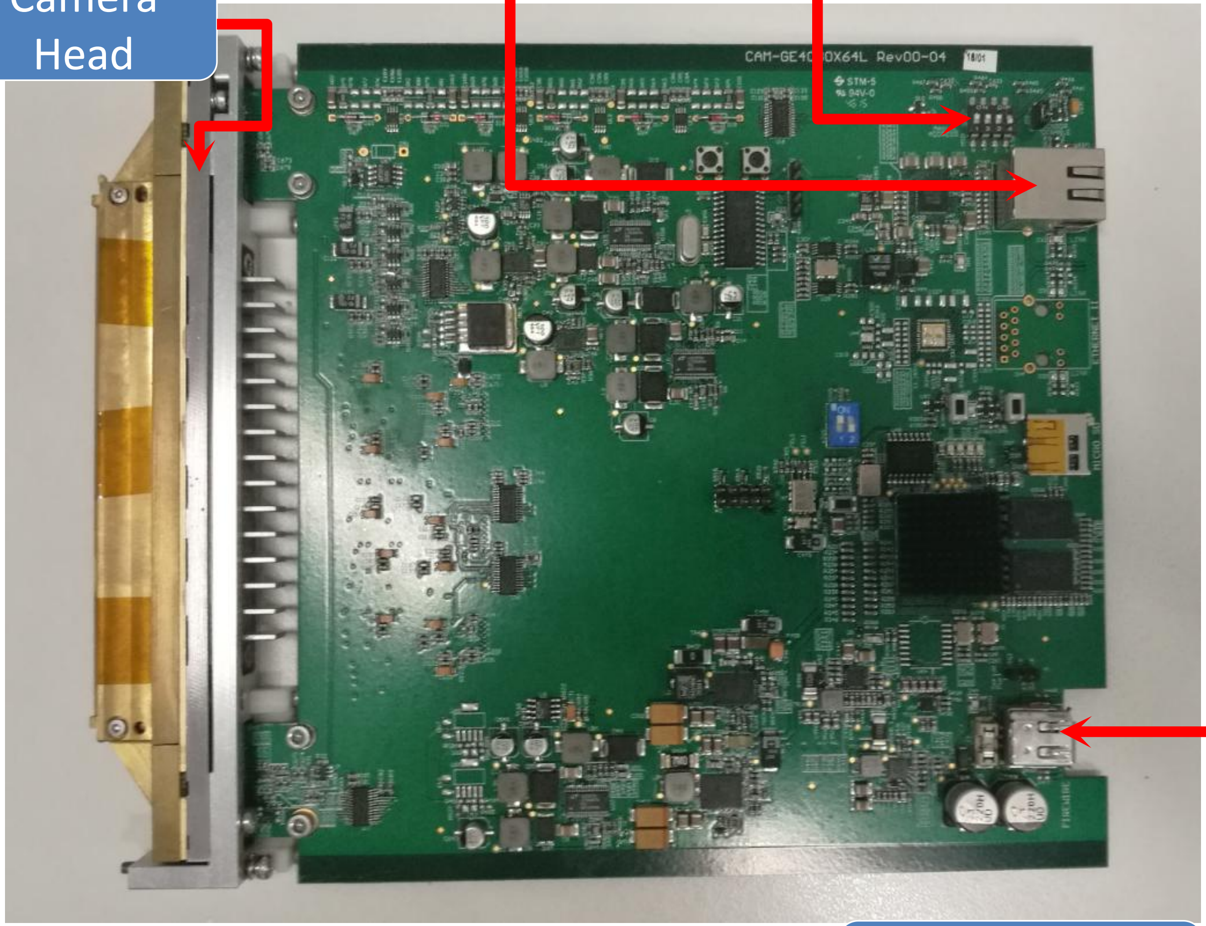
Camera Head

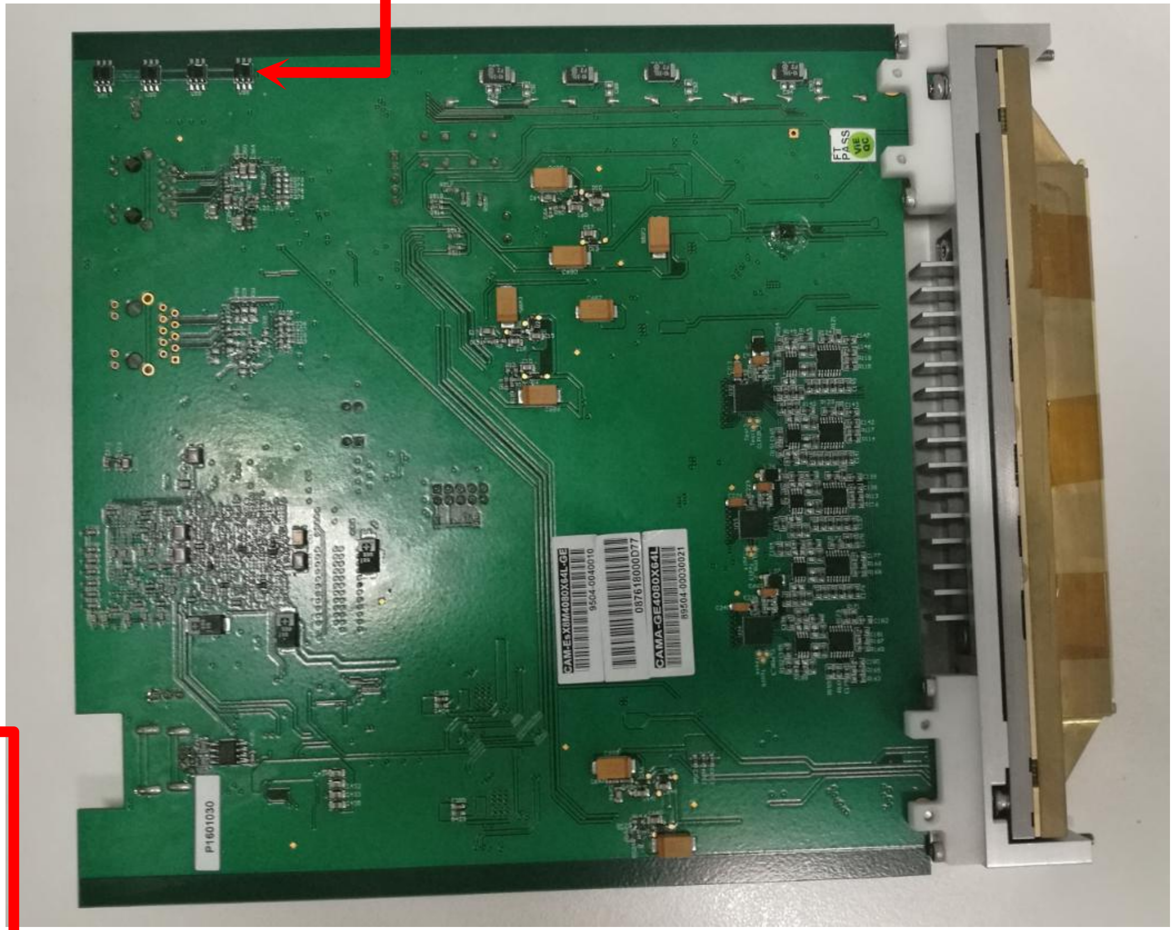
Connect to Gigabit Switch

ID Jumper

IR Sensor



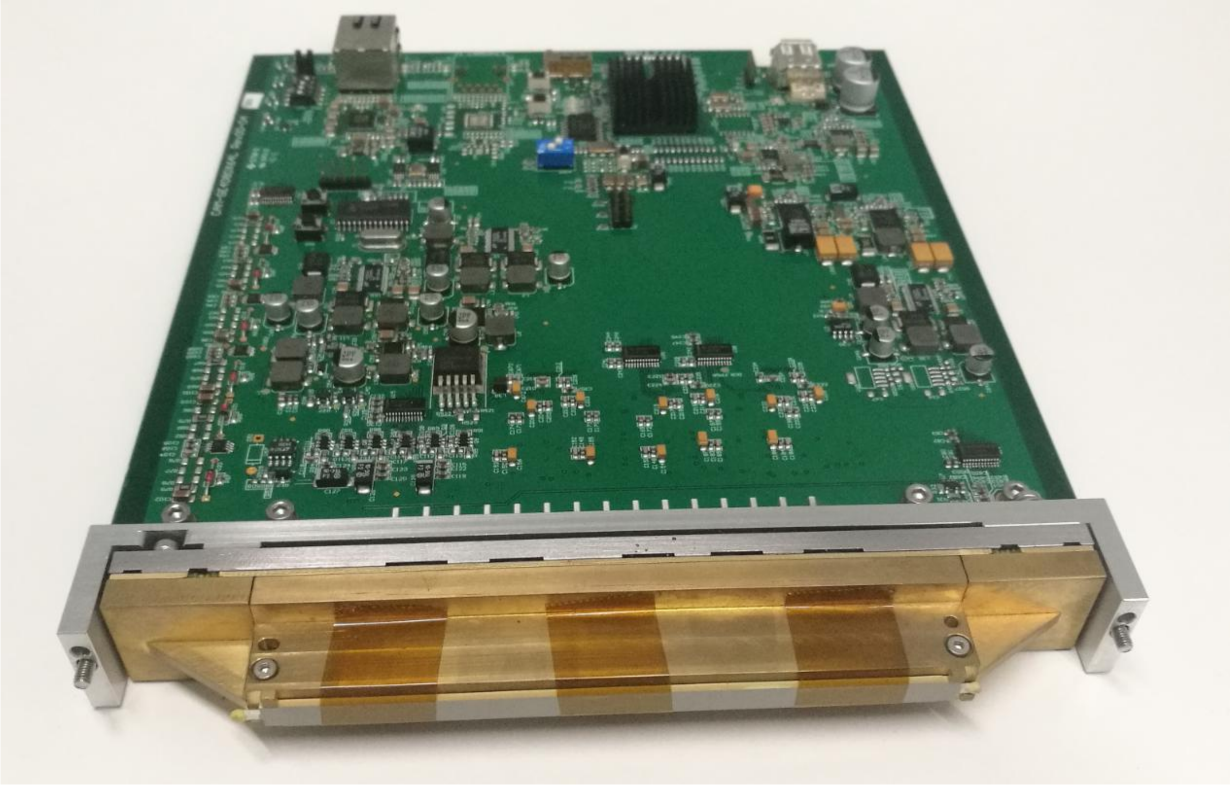
Top

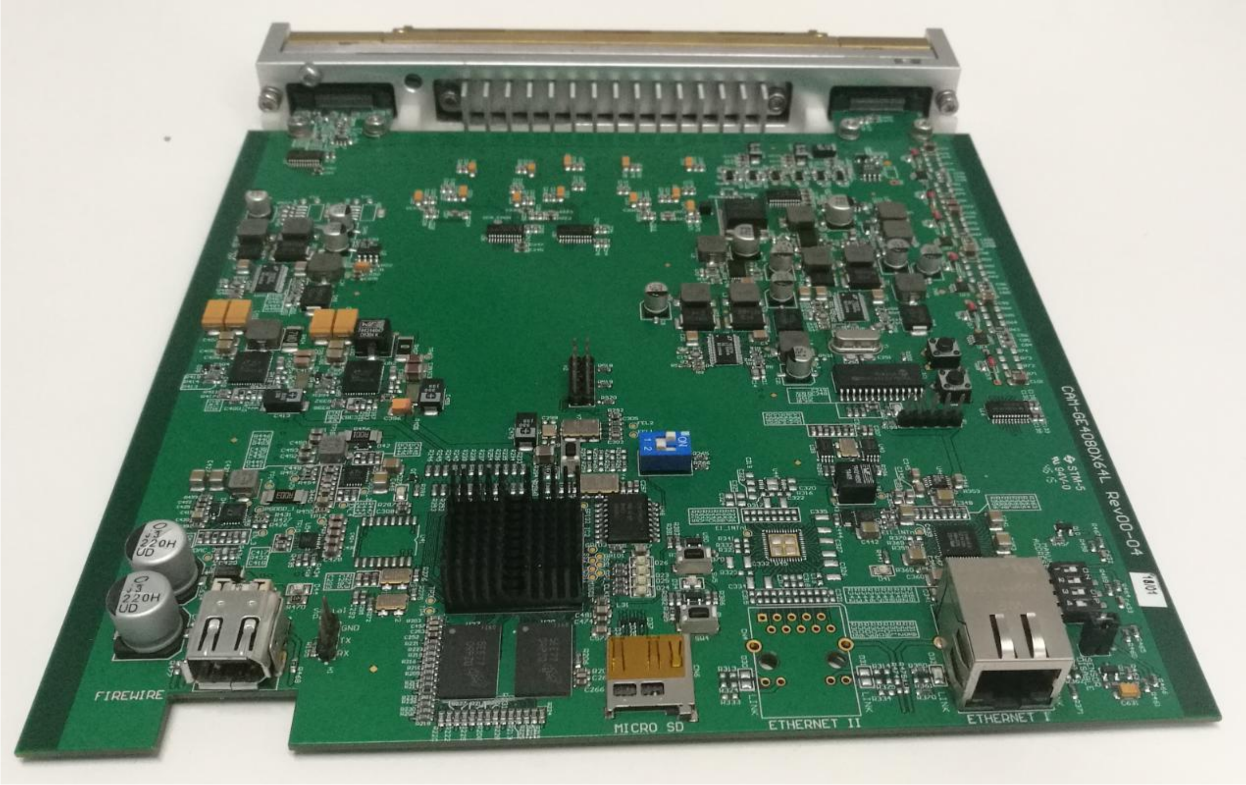


Bottom

Connect to Trigger Board

Hardware Overview - X-Ray Camera- Gen 3 Single Port





Hardware Overview - X-Ray Camera- Gen 3 Dual Port

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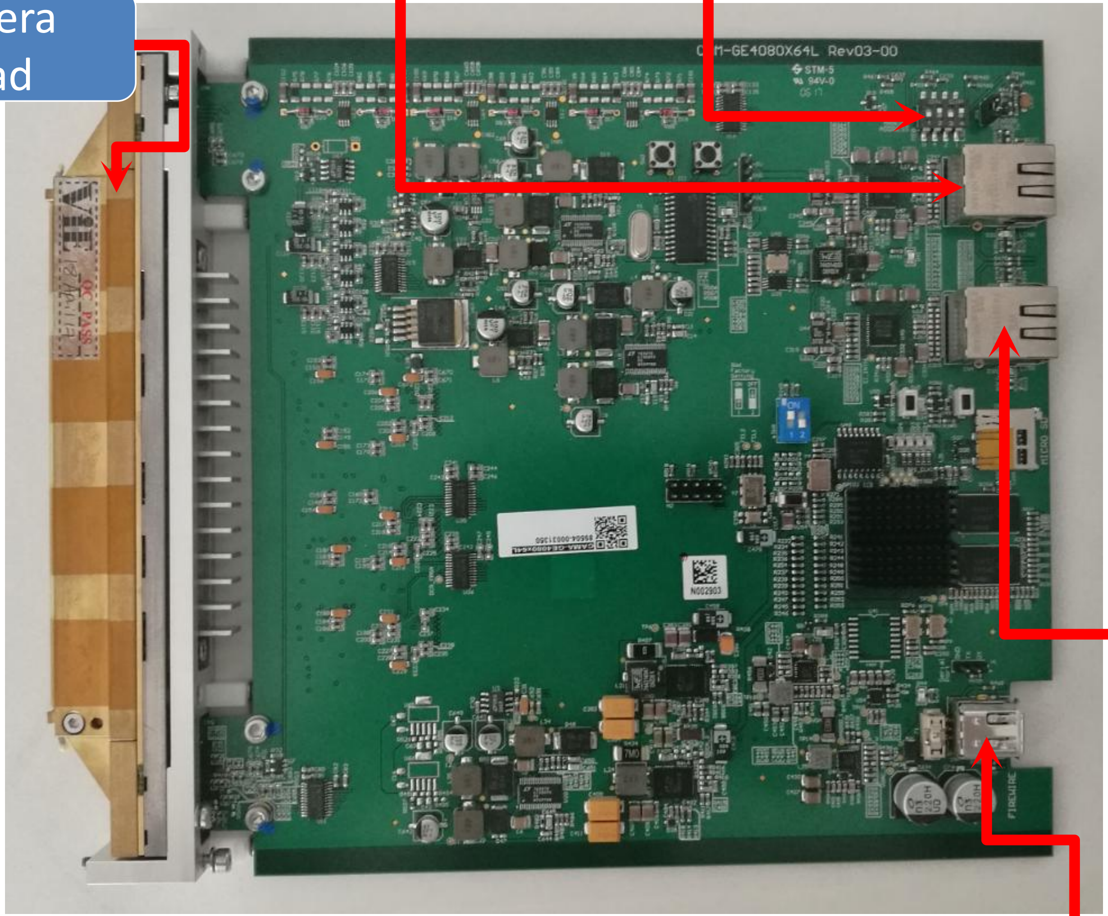
Connect to Gigabit Switch

ID Jumper

IR Sensor

Camera Head



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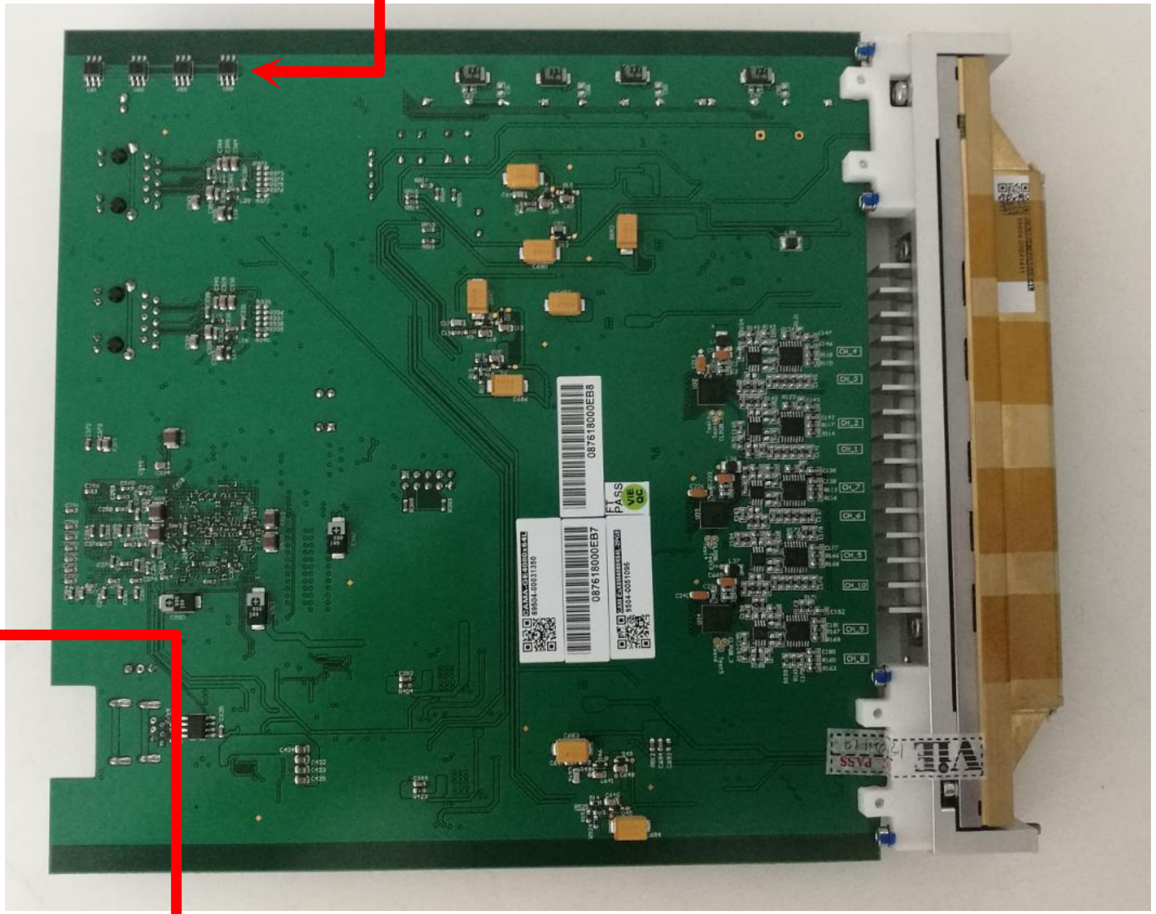

Bottom

Connect to Trigger Board

Not in used

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Hardware Overview - X-Ray Camera- Gen 3 Dual Port

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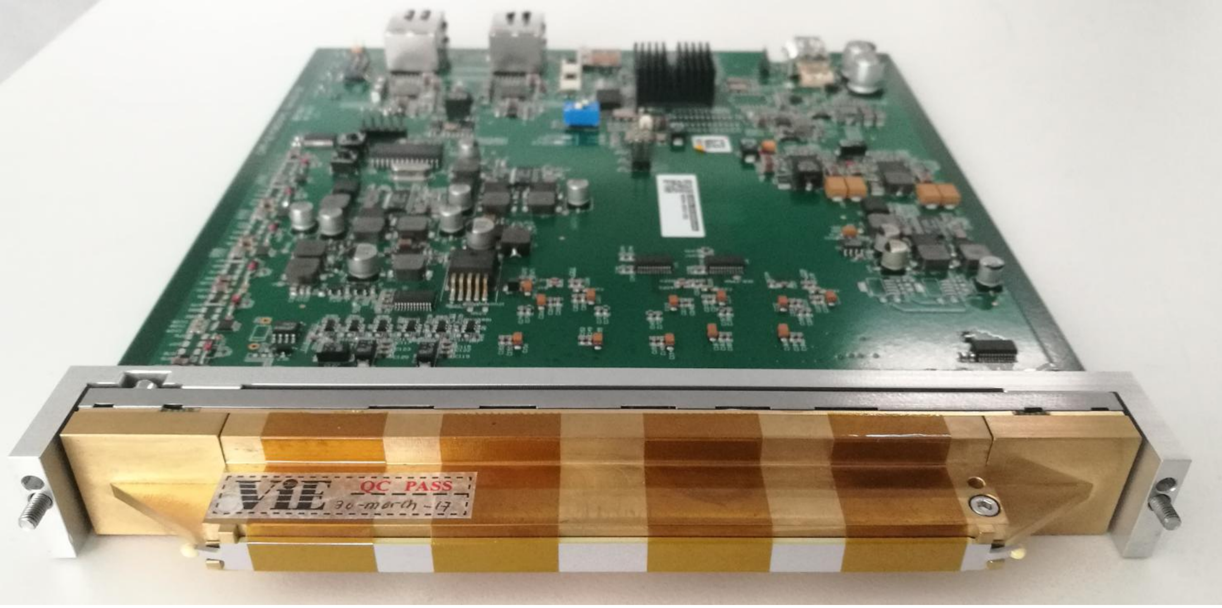
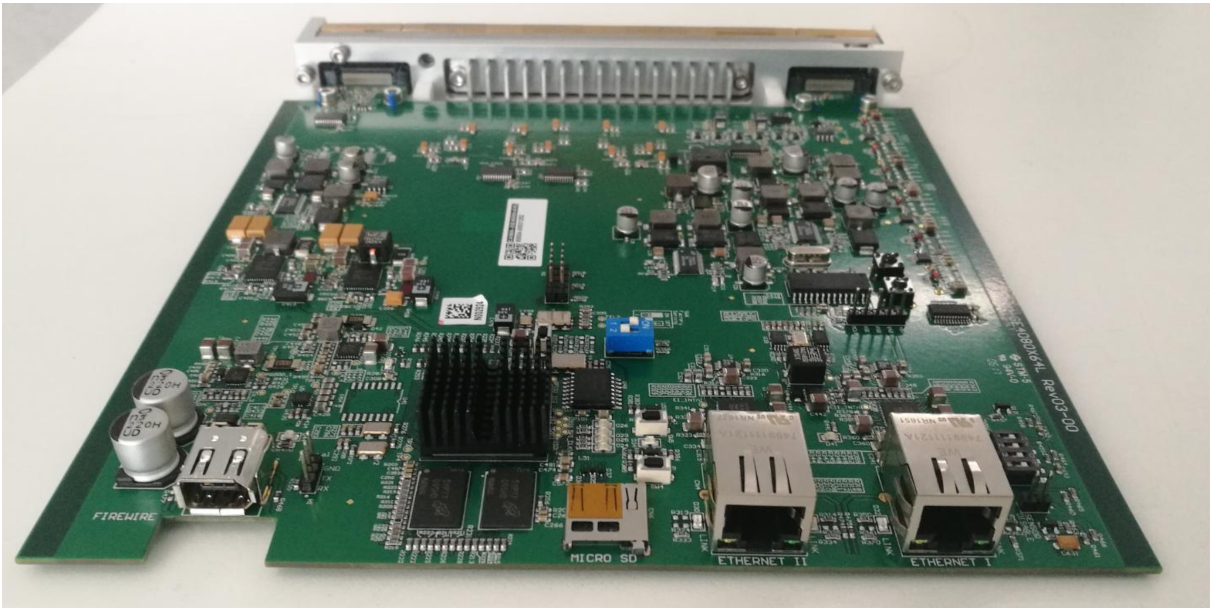
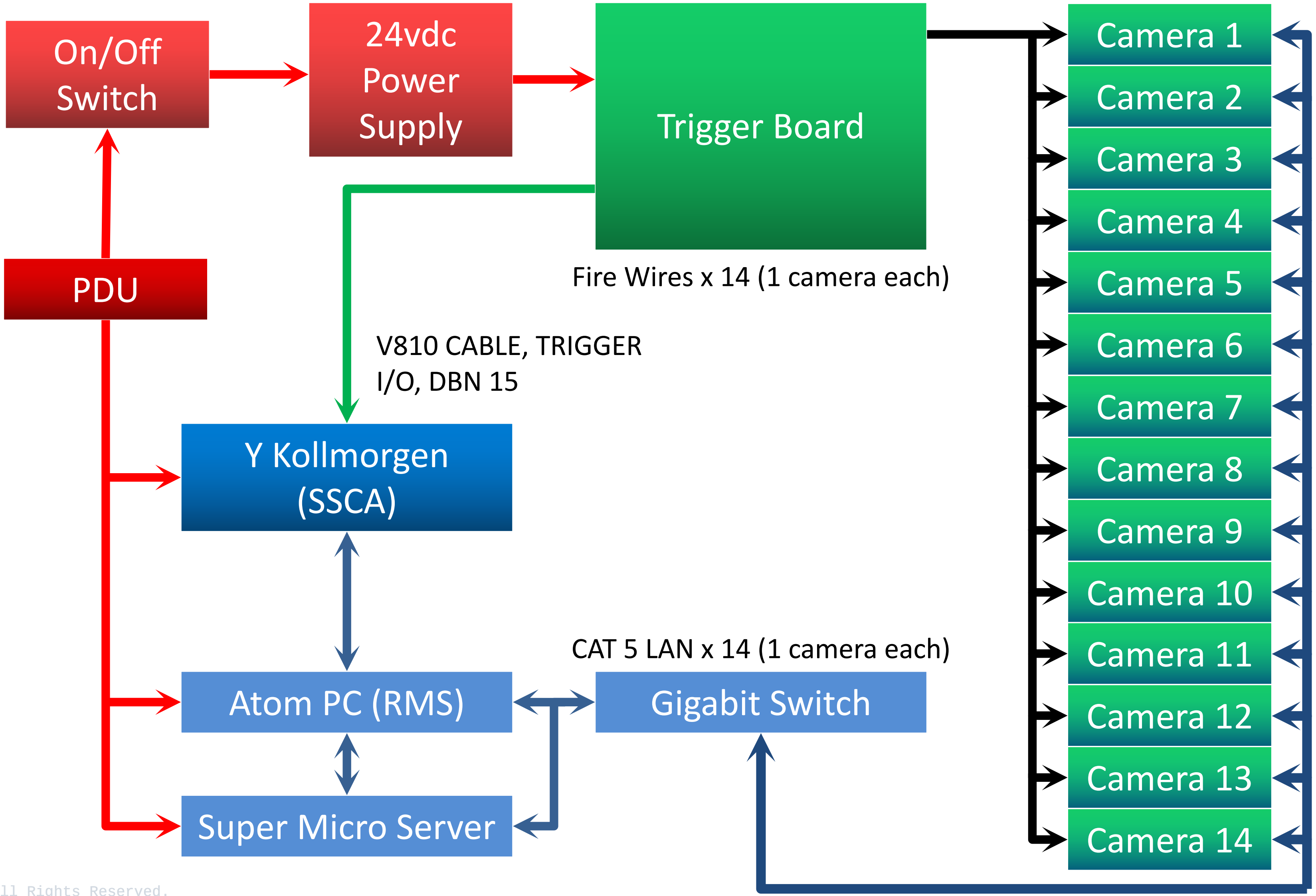
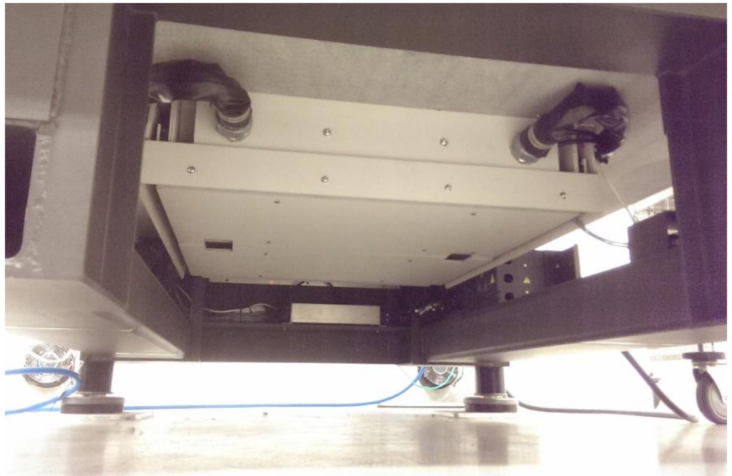




Camera Array Connection Diagram



Hardware Overview - X-Ray Camera Assembly



Y Axis Motion Hardware Overview

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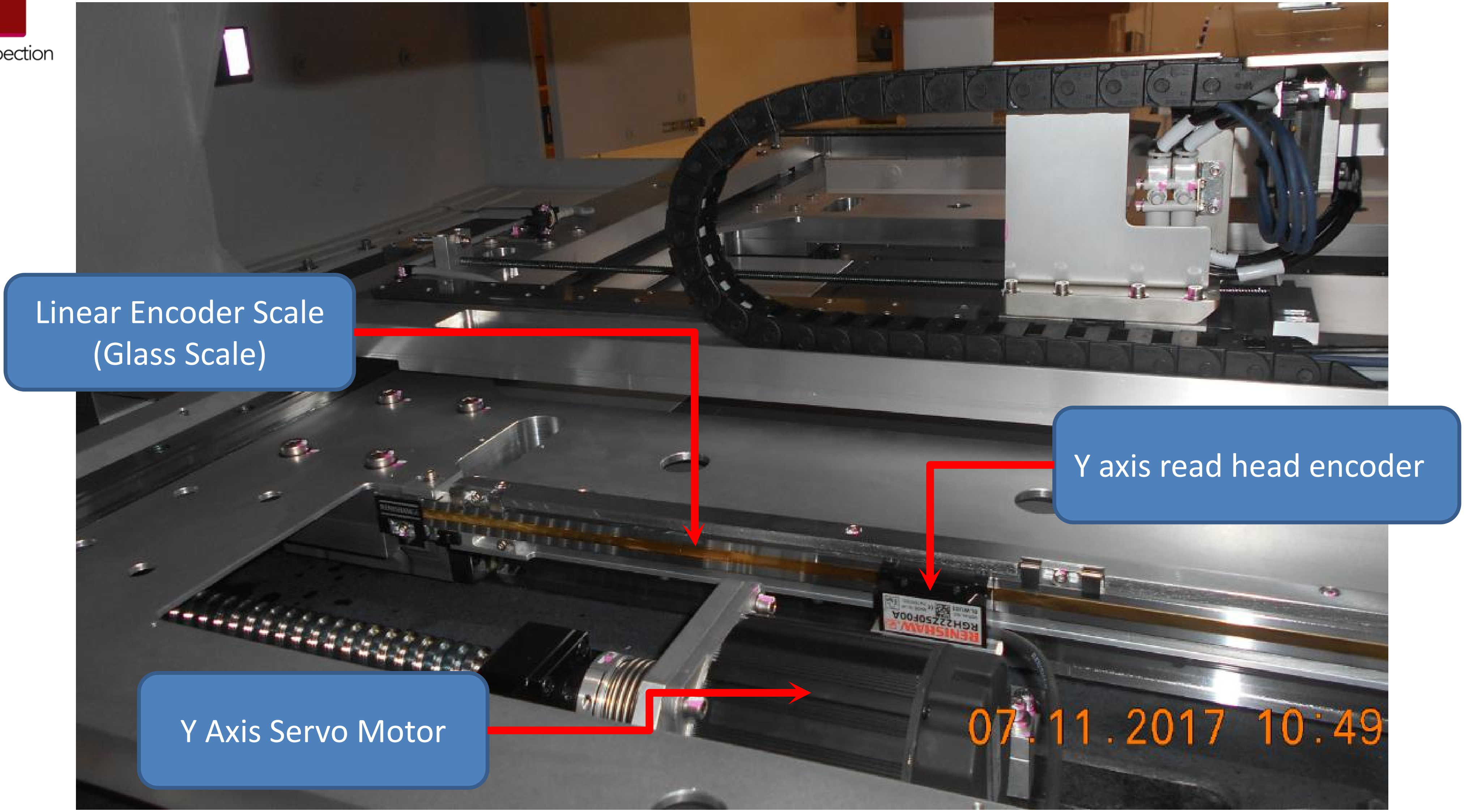
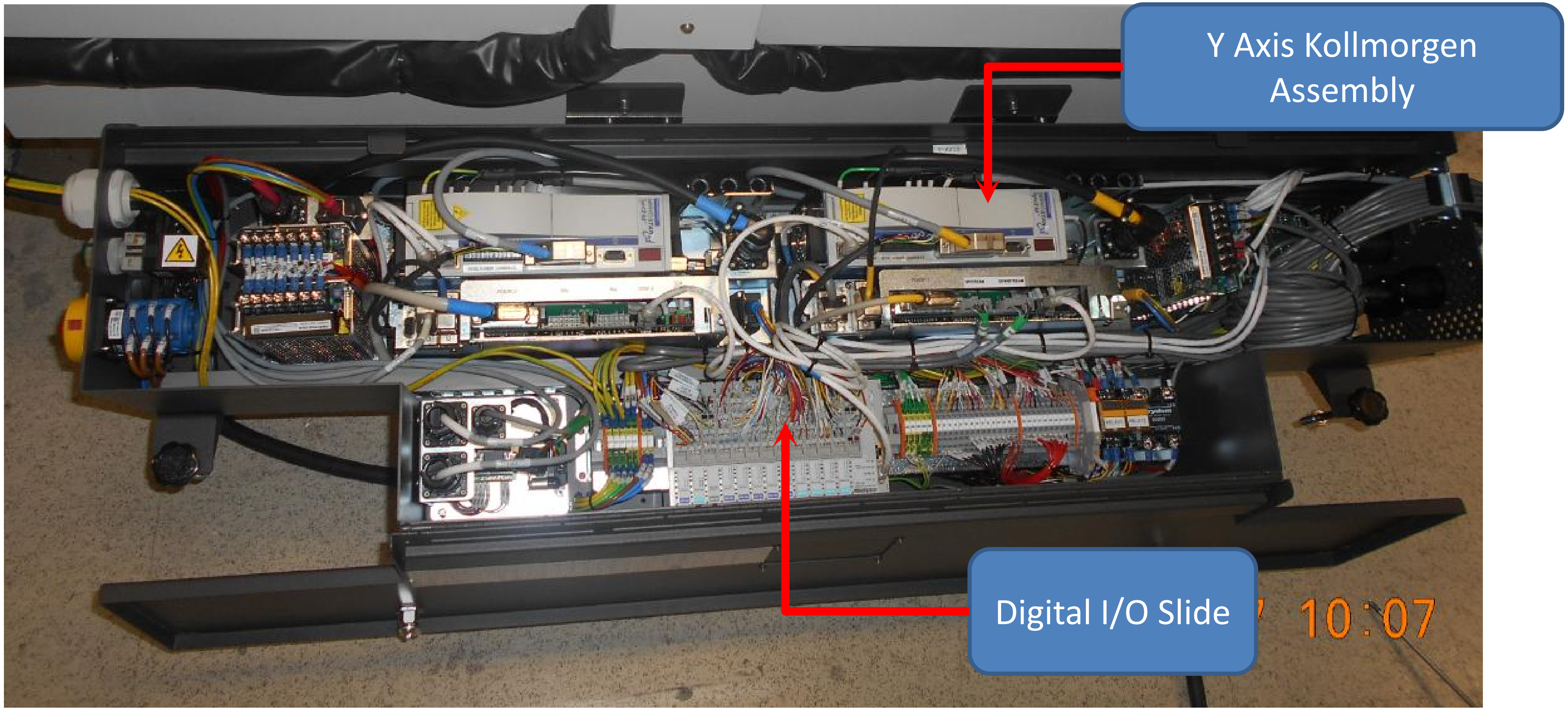


Y Axis Motion Hardware Overview



Y Axis Motion Hardware Overview

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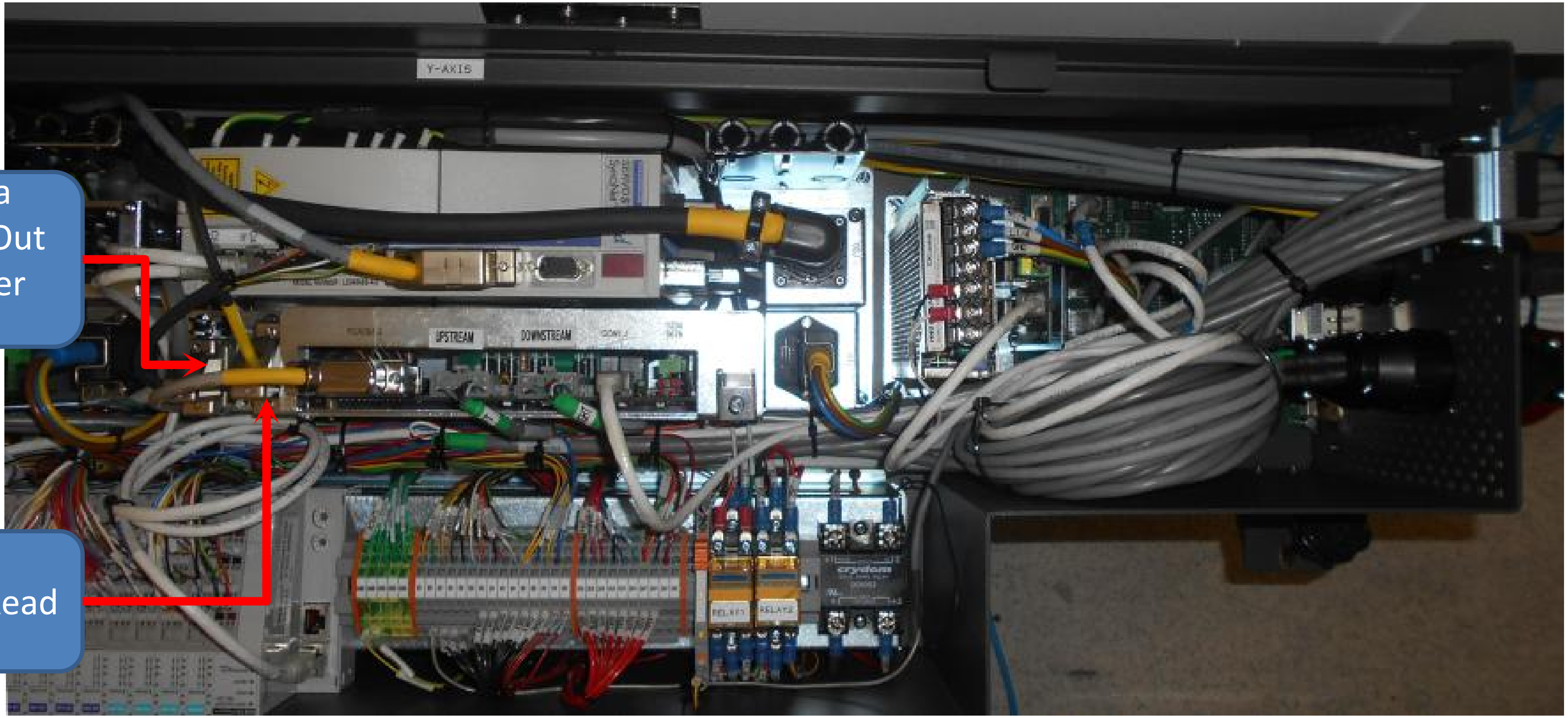
Camera Triggers Out to Trigger Board



Y-Axis Encoder Read Head

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...innovating vision

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Vice President & General Manager

GARY LEONG
Sales Director

周道南
Sales General Manager (China & Taiwan)

SEOW ZI YANG
Business Development Manager

FAHMI HELMI
Business Development Manager